

microsoft/referencesource: System.Data/System/Data/DataSet.cs

microsoft/referencesource: System.Data/System/Data/DataSet.cs - microsoft, GitHub.

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Content

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DataSet.cs

`microsoft/referencesource` at `main`, path `System.Data/System/Data/DataSet.cs` .

The citation points at line 323.

```
//-----
// <copyright file="DataSet.cs" company="Microsoft">
//     Copyright (c) Microsoft Corporation. All rights reserved.
// </copyright>
// <owner current="true" primary="true">Microsoft</owner>
// <owner current="true" primary="false">Microsoft</owner>
//-----

namespace System.Data {
    using System;
    using System.Text;
    using System.Runtime.Serialization;
    using System.ComponentModel;
    using System.Diagnostics;
    using System.Globalization;
    using System.IO;
    using System.Collections;
    using System.Collections.Generic;
    using System.Collections.Specialized;
    using System.Xml;
    using System.Xml.Serialization;
    using System.Xml.Serialization.Advanced;
    using System.Xml.Schema;
    using System.Runtime.Serialization.Formatters.Binary; //Binary Formatter
    using System.CodeDom;
    using System.CodeDom.Compiler;
    using System.Configuration;
    using System.Data.Common;
    using System.Runtime.Versioning;
    using System.Runtime.CompilerServices;

    /// <devdoc>
    ///     <para>
    ///         Represents an in-memory cache of data.
    ///     </para>
    /// </devdoc>
    [
        ResDescriptionAttribute(Res.DataSetDescr),
        DefaultProperty("DataSetName"),
        ToolboxItem("Microsoft.VSDesigner.Data.VS.DataSetToolboxItem, " +
AssemblyRef.MicrosoftVSDesigner),
        Designer("Microsoft.VSDesigner.Data.VS.DataSetDesigner, " +
AssemblyRef.MicrosoftVSDesigner),
        Serializable,
    ]
}
```

```

    XmlSchemaProvider("GetDataSetSchema"),
    XmlRoot("DataSet")
}

public class DataSet : MarshalByValueComponent, System.ComponentModel.IListSource,
IXmlSerializable, ISupportInitializeNotification, ISerializable {

    private DataViewManager defaultManager;

    // Public Collections
    private readonly DataTableCollection tableCollection;
    private readonly DataRelationCollection relationCollection;
    internal PropertyCollection extendedProperties = null;
    private string dataSetName = "NewDataSet";
    private string _datasetPrefix = String.Empty;
    internal string namespaceURI = string.Empty;
    private bool enforceConstraints = true;
    private const String KEY_XMLSCHEMA = "XmlSchema";
    private const String KEY_XMLDIFFGRAM = "XmlDiffGram";

    // globalization stuff
    private bool _caseSensitive;
    private CultureInfo _culture;
    private bool _cultureUserSet;

    // Internal definitions
    internal bool fInReadXml = false;
    internal bool fInLoadDiffgram = false;
    internal bool fTopLevelTable = false;
    internal bool fInitInProgress = false;
    internal bool fEnableCascading = true;
    internal bool fIsSchemaLoading = false;
    private bool fBoundToDocument; // for XmlDataDocument

    // Events
    private PropertyChangedEventHandler onPropertyChangingDelegate;
    private MergeFailedEventHandler onMergeFailed;
    private DataRowCreatedEventHandler onDataRowCreated; // Internal for
XmlDataDocument only
    private DataSetClearEventHandler onClearFunctionCalled; // Internal for
XmlDataDocument only
    private System.EventHandler onInitialized;

    internal readonly static DataTable[] zeroTables = new DataTable[0];

```

```

internal string mainTableName = "";

//default remoting format is XML
private SerializationFormat _remotingFormat = SerializationFormat.Xml;

private object _defaultViewManagerLock = new Object();

private static int _objectTypeCount; // Bid counter
private readonly int _objectID = System.Threading.Interlocked.Increment(ref
_objectTypeCount);
private static XmlSchemaComplexType schemaTypeForWSDL = null;

internal bool UseDataSetSchemaOnly; // UseDataSetSchemaOnly , for YUKON
internal bool UdtIsWrapped; // if UDT is wrapped , for YUKON

/// <devdoc>
/// <para>Initializes a new instance of the <see cref='System.Data.DataSet'/>
class.</para>
/// </devdoc>
public DataSet() {
    GC.SuppressFinalize(this);
    Bid.Trace("<ds.DataSet.DataSet|API> %d#\n", ObjectID); // others will call
this constr
    // Set default locale
    this.tableCollection = new DataTableCollection(this);
    this.relationCollection = new
DataRelationCollection.DataSetRelationCollection(this);
    _culture = CultureInfo.CurrentCulture; // Set default locale
}

/// <devdoc>
/// <para>Initializes a new instance of a <see cref='System.Data.DataSet'/>
/// class with the given name.</para>
/// </devdoc>
public DataSet(string dataSetName)
    : this()
{
    this.DataSetName = dataSetName;
}

[
DefaultValue(SerializationFormat.Xml)
]
public SerializationFormat RemotingFormat {

```

```

        get {
            return _remotingFormat;
        }
        set {
            if (value != SerializationFormat.Binary && value !=
SerializationFormat.Xml) {
                throw ExceptionBuilder.InvalidRemotingFormat(value);
            }
            _remotingFormat = value;
            // this property is inherited to DataTable from DataSet. So we set this
value to DataTable also
            for (int i = 0; i < Tables.Count; i++) {
                Tables[i].RemotingFormat = value;
            }
        }
    }

    [BrowsableAttribute(false)]
    [DesignerSerializationVisibility(DesignerSerializationVisibility.Hidden)]
    public virtual SchemaSerializationMode SchemaSerializationMode { //Typed DataSet
calls into this
        get {
            return SchemaSerializationMode.IncludeSchema;
        }
        set {
            if (value != SchemaSerializationMode.IncludeSchema) {
                throw ExceptionBuilder.CannotChangeSchemaSerializationMode();
            }
        }
    }

    //      Check whether the stream is binary serialized.
    // 'static' function that consumes SerializationInfo
    protected bool IsBinarySerialized(SerializationInfo info, StreamingContext
context) {// mainly for typed DS
        // our default remoting format is XML
        SerializationFormat remotingFormat = SerializationFormat.Xml;
        SerializationInfoEnumerator e = info.GetEnumerator();

        while (e.MoveNext()) {
            if (e.Name == "DataSet.RemotingFormat") {//DataSet.RemotingFormat does
not exist in V1/V1.1 versions
                remotingFormat = (SerializationFormat)e.Value;
                break;
            }
        }
    }

```

```

    }
}
return (remotingFormat == SerializationFormat.Binary);
}

//      Should Schema be included during Serialization
// 'static' function that consumes SerializationInfo
protected SchemaSerializationMode
DetermineSchemaSerializationMode(SerializationInfo info, StreamingContext context) {
//Typed DataSet calls into this
    SchemaSerializationMode schemaSerializationMode =
SchemaSerializationMode.IncludeSchema;
    SerializationInfoEnumerator e = info.GetEnumerator();

    while (e.MoveNext()) {
        if (e.Name == "SchemaSerializationMode.DataSet") {
//SchemaSerializationMode.DataSet does not exist in V1/V1.1 versions
            schemaSerializationMode = (SchemaSerializationMode)e.Value;
            break;
        }
    }
    return schemaSerializationMode;
}

protected SchemaSerializationMode DetermineSchemaSerializationMode(XmlReader
reader) { //Typed DataSet calls into this
    SchemaSerializationMode schemaSerializationMode =
SchemaSerializationMode.IncludeSchema;
    reader.MoveToContent();
    if (reader.NodeType == XmlNodeType.Element) {
        if (reader.HasAttributes) {
            string attribValue =
reader.GetAttribute(Keywords.MSD_SCHEMASERIALIZATIONMODE, Keywords.MSDNS);
            if ((String.Compare(attribValue, Keywords.MSD_EXCLUDESCHEMA,
StringComparison.OrdinalIgnoreCase) == 0)) {
                schemaSerializationMode = SchemaSerializationMode.ExcludeSchema;
            }
            else if ((String.Compare(attribValue, Keywords.MSD_INCLUDESCHEMA,
StringComparison.OrdinalIgnoreCase) == 0)) {
                schemaSerializationMode = SchemaSerializationMode.IncludeSchema;
            }
            else if (attribValue != null) { // if attrib does not exist, then
don't throw
                throw

```

```

ExceptionBuilder.InvalidSchemaSerializationMode(typeof(SchemaSerializationMode),
attribValue);
    }
}
}
return schemaSerializationMode;
}

//      Deserialize all the tables data of the dataset from binary/xml stream.
// 'instance' method that consumes SerializationInfo
protected void GetSerializationData(SerializationInfo info, StreamingContext
context) { // mainly for typed DS
    SerializationFormat remotingFormat = SerializationFormat.Xml;
    SerializationInfoEnumerator e = info.GetEnumerator();

    while (e.MoveNext()) {
        if (e.Name == "DataSet.RemotingFormat") { //DataSet.RemotingFormat does
not exist in V1/V1.1 versions
            remotingFormat = (SerializationFormat)e.Value;
            break;
        }
    }
    DeserializeDataSetData(info, context, remotingFormat);
}

//      Deserialize all the tables schema and data of the dataset from binary/xml
stream.
protected DataSet(SerializationInfo info, StreamingContext context)
: this(info, context, true) {
}
protected DataSet(SerializationInfo info, StreamingContext context, bool
ConstructSchema)
: this() {
    SerializationFormat remotingFormat = SerializationFormat.Xml;
    SchemaSerializationMode schemaSerializationMode =
SchemaSerializationMode.IncludeSchema;
    SerializationInfoEnumerator e = info.GetEnumerator();

    while (e.MoveNext()) {
        switch (e.Name) {
            case "DataSet.RemotingFormat": //DataSet.RemotingFormat does not
exist in V1/V1.1 versions

```

```

        remotingFormat = (SerializationFormat)e.Value;
        break;
        case "SchemaSerializationMode.DataSet":
//SchemaSerializationMode.DataSet does not exist in V1/V1.1 versions
        schemaSerializationMode = (SchemaSerializationMode)e.Value;
        break;
    }
}

if (schemaSerializationMode == SchemaSerializationMode.ExcludeSchema) {
    InitializeDerivedDataSet();
}

// adding back this check will fix typed dataset XML remoting, but we have to
fix case that
// a class inherits from DataSet and just relies on DataSet to deserialize
(see SQL BU DT 374717)
// to fix that case also, we need to add a flag and add it to below check so
return (no-op) will be
// conditional (flag needs to be set in TypedDataSet
if (remotingFormat == SerializationFormat.Xml && !ConstructSchema /* &&
this.GetType() != typeof(DataSet)*/) {
    return; //For typed dataset xml remoting, this is a no-op
}
DeserializeDataSet(info, context, remotingFormat, schemaSerializationMode);
}

```

```

[System.Security.Permissions.SecurityPermissionAttribute(System.Security.Permissions.SecurityAction.LinkDemand, Flags =
System.Security.Permissions.SecurityPermissionFlag.SerializationFormatter)]

```

```

    public virtual void GetObjectData(SerializationInfo info, StreamingContext
context) {

```

```

        SerializationFormat remotingFormat = RemotingFormat;
        SerializeDataSet(info, context, remotingFormat);
    }

```

```

//      Deserialize all the tables data of the dataset from binary/xml stream.

```

```

protected virtual void InitializeDerivedDataSet() {
    return;
}

```

```

        //      Serialize all the tables.
        private void SerializeDataSet(SerializationInfo info, StreamingContext context,
        SerializationFormat remotingFormat) {
            Debug.Assert(info != null);
            info.AddValue("DataSet.RemotingVersion", new Version(2, 0));

            // SqlHotFix 299, SerializationFormat enumeration types don't exist in V1.1
        SP1
            if (SerializationFormat.Xml != remotingFormat) {
                info.AddValue("DataSet.RemotingFormat", remotingFormat);
            }

            // SqlHotFix 299, SchemaSerializationMode enumeration types don't exist in
        V1.1 SP1
            if (SchemaSerializationMode.IncludeSchema != SchemaSerializationMode) {
                //SkipSchemaDuringSerialization
                info.AddValue("SchemaSerializationMode.DataSet",
        SchemaSerializationMode);
            }

            if (remotingFormat != SerializationFormat.Xml) {
                if (SchemaSerializationMode == SchemaSerializationMode.IncludeSchema) {

                    //DataSet public state properties
                    SerializeDataSetProperties(info, context);

                    //Tables Count
                    info.AddValue("DataSet.Tables.Count", Tables.Count);

                    //Tables, Columns, Rows
                    for (int i = 0; i < Tables.Count; i++) {
                        BinaryFormatter bf = new BinaryFormatter(null, new
        StreamingContext(context.State, false));
                        MemoryStream memStream = new MemoryStream();
                        bf.Serialize(memStream, Tables[i]);
                        memStream.Position = 0;
                        info.AddValue(String.Format(CultureInfo.InvariantCulture,
        "DataSet.Tables_{0}", i), memStream.GetBuffer());
                    }

                    //Constraints
                    for (int i = 0; i < Tables.Count; i++) {
                        Tables[i].SerializeConstraints(info, context, i, true);
                    }
                }
            }
        }
    }

```

```

        //Relations
        SerializeRelations(info, context);

        //Expression Columns
        for (int i = 0; i < Tables.Count; i++) {
            Tables[i].SerializeExpressionColumns(info, context, i);
        }
    }
    else {
        //Serialize DataSet public properties.
        SerializeDataSetProperties(info, context);
    }
    //Rows
    for (int i = 0; i < Tables.Count; i++) {
        Tables[i].SerializeTableData(info, context, i);
    }
} else { // old behaviour

    String strSchema = this.GetXmlSchemaForRemoting(null);

    String strData = null;
    info.AddValue(KEY_XMLSCHEMA, strSchema);

    StringBuilder strBuilder;
    strBuilder = new StringBuilder(EstimatedXmlStringSize() * 2);
    StringWriter strWriter = new StringWriter(strBuilder,
CultureInfo.InvariantCulture);
    XmlTextWriter w = new XmlTextWriter(strWriter);
    WriteXml(w, XmlWriteMode.DiffGram);
    strData = strWriter.ToString();
    info.AddValue(KEY_XMLDIFFGRAM, strData);
}
}

//      Deserialize all the tables - marked internal so that DataTable can call
into this
    internal void DeserializeDataSet(SerializationInfo info, StreamingContext
context, SerializationFormat remotingFormat, SchemaSerializationMode
schemaSerializationMode) {
        // deserialize schema
        DeserializeDataSetSchema(info, context, remotingFormat,
schemaSerializationMode);
        // deserialize data

```

```

        DeserializeDataSetData(info, context, remotingFormat);
    }

    // Deserialize schema.
    private void DeserializeDataSetSchema(SerializationInfo info, StreamingContext
context, SerializationFormat remotingFormat, SchemaSerializationMode
schemaSerializationMode) {
        if (remotingFormat != SerializationFormat.Xml) {
            if (schemaSerializationMode == SchemaSerializationMode.IncludeSchema) {
                //DataSet public state properties
                DeserializeDataSetProperties(info, context);

                //Tables Count
                int tableCount = info.GetInt32("DataSet.Tables.Count");

                //Tables, Columns, Rows
                for (int i = 0; i < tableCount; i++) {
                    Byte[] buffer =
(Byte[][])info.GetValue(String.Format(CultureInfo.InvariantCulture, "DataSet.Tables_{0}",
i), typeof(Byte[]));

                    MemoryStream memStream = new MemoryStream(buffer);
                    memStream.Position = 0;
                    BinaryFormatter bf = new BinaryFormatter(null, new
StreamingContext(context.State, false));
                    DataTable dt = (DataTable)bf.Deserialize(memStream);
                    Tables.Add(dt);
                }

                //Constraints
                for (int i = 0; i < tableCount; i++) {
                    Tables[i].DeserializeConstraints(info, context, /* table index
*/i, /* serialize all constraints */ true); //
                }

                //Relations
                DeserializeRelations(info, context);

                //Expression Columns
                for (int i = 0; i < tableCount; i++) {
                    Tables[i].DeserializeExpressionColumns(info, context, i);
                }
            } else {
                //DeSerialize DataSet public properties.[Locale, CaseSensitive and
EnforceConstraints]

```

```

        DeserializeDataSetProperties(info, context);
    }
} else {
    string strSchema = (String)info.GetValue(KEY_XMLSCHEMA,
typeof(System.String));

    if (strSchema != null) {
        this.ReadXmlSchema(new XmlTextReader(new StringReader(strSchema)),
true);
    }
}

//      Deserialize all data.
private void DeserializeDataSetData(SerializationInfo info, StreamingContext
context, SerializationFormat remotingFormat) {
    if (remotingFormat != SerializationFormat.Xml) {
        for (int i = 0; i < Tables.Count; i++) {
            Tables[i].DeserializeTableData(info, context, i);
        }
    }
    else {
        string strData = (String)info.GetValue(KEY_XMLDIFFGRAM,
typeof(System.String));

        if (strData != null) {
            this.ReadXml(new XmlTextReader(new StringReader(strData)),
XmlReadMode.DiffGram);
        }
    }
}

//      Serialize just the dataset properties
private void SerializeDataSetProperties(SerializationInfo info, StreamingContext
context) {
    //DataSet basic properties
    info.AddValue("DataSet.DataSetName", DataSetName);
    info.AddValue("DataSet.Namespace", Namespace);
    info.AddValue("DataSet.Prefix", Prefix);
    //DataSet runtime properties
    info.AddValue("DataSet.CaseSensitive", CaseSensitive);
    info.AddValue("DataSet.LocaleLCID", Locale.LCID);
    info.AddValue("DataSet.EnforceConstraints", EnforceConstraints);
}

```

```

        //ExtendedProperties
        info.AddValue("DataSet.ExtendedProperties", ExtendedProperties);
    }

    //      DeSerialize dataset properties
    private void DeserializeDataSetProperties(SerializationInfo info,
StreamingContext context) {
        //DataSet basic properties
        dataSetName = info.GetString("DataSet.DataSetName");
        namespaceURI = info.GetString("DataSet.Namespace");
        _datasetPrefix = info.GetString("DataSet.Prefix");
        //DataSet runtime properties
        _caseSensitive = info.GetBoolean("DataSet.CaseSensitive");
        int lcid = (int)info.GetValue("DataSet.LocaleLCID", typeof(int));
        _culture = new CultureInfo(lcid);
        _cultureUserSet = true;
        enforceConstraints = info.GetBoolean("DataSet.EnforceConstraints");

        //ExtendedProperties
        extendedProperties =
(PropertyCollection)info.GetValue("DataSet.ExtendedProperties",
typeof(PropertyCollection));
    }

    //      Gets relation info from the dataset.
    //      ***Schema for Serializing ArrayList of Relations***
    //      Relations -> [relationName]->[parentTableIndex, parentcolumnIndexes]->
[childTableIndex, childColumnIndexes]->[Nested]->[extendedProperties]
    private void SerializeRelations(SerializationInfo info, StreamingContext context)
    {
        ArrayList relationList = new ArrayList();

        foreach (DataRelation rel in Relations) {
            int[] parentInfo = new int[rel.ParentColumns.Length + 1];

            parentInfo[0] = Tables.IndexOf(rel.ParentTable);
            for (int j = 1; j < parentInfo.Length; j++) {
                parentInfo[j] = rel.ParentColumns[j - 1].Ordinal;
            }

            int[] childInfo = new int[rel.ChildColumns.Length + 1];
            childInfo[0] = Tables.IndexOf(rel.ChildTable);

```

```

        for (int j = 1; j < childInfo.Length; j++) {
            childInfo[j] = rel.ChildColumns[j - 1].Ordinal;
        }

        ArrayList list = new ArrayList();
        list.Add(rel.RelationName);
        list.Add(parentInfo);
        list.Add(childInfo);
        list.Add(rel.Nested);
        list.Add(rel.extendedProperties);

        relationList.Add(list);
    }
    info.AddValue("DataSet.Relations", relationList);
}

/*
    Adds relations to the dataset.
    ***Schema for Serializing ArrayList of Relations***
    Relations -> [relationName]->[parentTableIndex, parentcolumnIndexes]->
    [childTableIndex, childColumnIndexes]->[Nested]->[extendedProperties]
*/
private void DeserializeRelations(SerializationInfo info, StreamingContext
context) {
    ArrayList relationList = (ArrayList)info.GetValue("DataSet.Relations",
typeof(ArrayList));

    foreach (ArrayList list in relationList) {
        string relationName = (string)list[0];
        int[] parentInfo = (int[])list[1];
        int[] childInfo = (int[])list[2];
        bool isNested = (bool)list[3];
        PropertyCollection extendedProperties = (PropertyCollection)list[4];

        //ParentKey Columns.
        DataColumn[] parentkeyColumns = new DataColumn[parentInfo.Length - 1];
        for (int i = 0; i < parentkeyColumns.Length; i++) {
            parentkeyColumns[i] = Tables[parentInfo[0]].Columns[parentInfo[i +
1]];
        }

        //ChildKey Columns.
        DataColumn[] childkeyColumns = new DataColumn[childInfo.Length - 1];
        for (int i = 0; i < childkeyColumns.Length; i++) {

```

```

        childkeyColumns[i] = Tables[childInfo[0]].Columns[childInfo[i + 1]];
    }

    //Create the Relation, without any constraints[Assumption: The
constraints are added earlier than the relations]
    DataRelation rel = new DataRelation(relationName, parentkeyColumns,
childkeyColumns, false);
    rel.CheckMultipleNested = false; // disable the check for multiple nested
parent

    rel.Nested = isNested;
    rel.extendedProperties = extendedProperties;

    Relations.Add(rel);
    rel.CheckMultipleNested = true; // enable the check for multiple nested
parent
    }
}

internal void FailedEnableConstraints() {
    this.EnforceConstraints = false;
    throw ExceptionBuilder.EnforceConstraint();
}

/// <devdoc>
///     <para>
///         Gets or sets a value indicating whether string
///         comparisons within <see cref='System.Data.DataTable' />
///         objects are
///         case-sensitive.
///     </para>
/// </devdoc>
[
    ResCategoryAttribute(Res.DataCategory_Data),
    DefaultValue(false),
    ResDescriptionAttribute(Res.DataSetCaseSensitiveDescr)
]
public bool CaseSensitive {
    get {
        return _caseSensitive;
    }
    set {
        if (_caseSensitive != value)
        {
            bool oldValue = _caseSensitive;

```

```

        _caseSensitive = value;

        if (!ValidateCaseConstraint()) {
            _caseSensitive = oldValue;
            throw ExceptionBuilder.CannotChangeCaseLocale();
        }

        foreach (DataTable table in Tables) {
            table.SetCaseSensitiveValue(value, false, true);
        }
    }
}

bool System.ComponentModel.IListSource.ContainsListCollection {
    get {
        return true;
    }
}

/// <devdoc>
/// <para>Gets a custom view of the data contained by the <see
cref='System.Data.DataSet'/> , one
///     that allows filtering, searching, and navigating through the custom data
view.</para>
/// </devdoc>
[Browsable(false), ResDescriptionAttribute(Res.DataSetDefaultViewDescr)]
public DataViewManager DefaultViewManager {
    get {
        if (defaultViewManager == null) {
            lock (_defaultViewManagerLock) {
                if (defaultViewManager == null) {
                    defaultViewManager = new DataViewManager(this, true);
                }
            }
        }
        return defaultViewManager;
    }
}

/// <devdoc>
/// <para>Gets or sets a value indicating whether constraint rules are
followed when
///     attempting any update operation.</para>

```

```

/// </devdoc>
[DefaultValue(true), ResDescriptionAttribute(Res.DataSetEnforceConstraintsDescr)]
public bool EnforceConstraints {
    get {
        return enforceConstraints;
    }
    set {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.set_EnforceConstraints|API> %d#,
%d{bool}\n", ObjectID, value);
        try {
            if (this.enforceConstraints != value) {
                if (value)
                    EnableConstraints();
                this.enforceConstraints = value;
            }
        }
        finally {
            Bid.ScopeLeave(ref hscp);
        }
    }
}

internal void RestoreEnforceConstraints(bool value) {
    this.enforceConstraints = value;
}

internal void EnableConstraints()
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.EnableConstraints|INFO> %d#\n",
ObjectID);
    try {
        bool errors = false;
        for (ConstraintEnumerator constraints = new ConstraintEnumerator(this);
constraints.Next(); ) {
            Constraint constraint = (Constraint)constraints.GetConstraint();
            errors |= constraint.IsConstraintViolated();
        }

        foreach (DataTable table in Tables) {
            foreach (DataColumn column in table.Columns) {
                if (!column.AllowDBNull) {
                    errors |= column.IsNotAllowDBNullViolated();
                }
            }
        }
    }
}

```

```

        }
        if (column.MaxLength >= 0) {
            errors |= column.IsMaxLengthViolated();
        }
    }
}

if (errors)
    this.FailedEnableConstraints();
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

/// <devdoc>
///     <para>Gets or
///         sets the name of this <see cref='System.Data.DataSet'/> .</para>
/// </devdoc>
[
    ResCategoryAttribute(Res.DataCategory_Data),
    DefaultValue(""),
    ResDescriptionAttribute(Res.DataSetDataSetNameDescr)
]
public string DataSetName {
    get {
        return dataSetName;
    }
    set {
        Bid.Trace("<ds.DataSet.set_DataSetName|API> %d#, '%ls'\n", ObjectID,
value);

        if (value != dataSetName) {
            if (value == null || value.Length == 0)
                throw ExceptionBuilder.SetDataSetNameToEmpty();
            DataTable conflicting = Tables[value, Namespace];
            if ((conflicting != null) && (!conflicting.fNestedInDataset))
                throw ExceptionBuilder.SetDataSetNameConflicting(value);
            RaisePropertyChanging("DataSetName");
            this.dataSetName = value;
        }
    }
}

/// <devdoc>

```

```

    /// </devdoc>
    [
    DefaultValue(""),
    ResCategoryAttribute(Res.DataCategory_Data),
    ResDescriptionAttribute(Res.DataSetNamespaceDescr)
    ]
    public string Namespace {
        get {
            return namespaceURI;
        }
        set {
            Bid.Trace("<ds.DataSet.set_Namespace|API> %d#, '%ls'\n", ObjectID,
value);

            if (value == null)
                value = String.Empty;
            if (value != namespaceURI) {
                RaisePropertyChanging("Namespace");
                foreach (DataTable dt in Tables) {
                    if (dt.tableNamespace != null)
                        continue;
                    if ((dt.NestedParentRelations.Length == 0) ||
                        (dt.NestedParentRelations.Length == 1 &&
dt.NestedParentRelations[0].ChildTable == dt)) {
                        // dt.SelfNestedWithOneRelation) {
// this is wrong bug it was previous behavior
                        if (Tables.Contains(dt.TableName, value, false, true))
                            throw ExceptionBuilder.DuplicateTableName2(dt.TableName,
value);

                        dt.CheckCascadingNamespaceConflict(value);
                        dt.DoRaiseNamespaceChange();
                    }
                }
                namespaceURI = value;

                if (Common.ADP.IsEmpty(value))
                    _datasetPrefix = String.Empty;
            }
        }
    }

    /// <devdoc>
    ///     <para>[To be supplied.]</para>
    /// </devdoc>

```

```

[
DefaultValue(""),
ResCategoryAttribute(Res.DataCategory_Data),
ResDescriptionAttribute(Res.DataSetPrefixDescr)
]
public string Prefix {
    get { return _datasetPrefix; }
    set {
        if (value == null)
            value = String.Empty;

        if ((XmlConvert.DecodeName(value) == value) &&
            (XmlConvert.EncodeName(value) != value))
            throw ExceptionBuilder.InvalidPrefix(value);

        if (value != _datasetPrefix) {
            RaisePropertyChanging("Prefix");
            _datasetPrefix = value;
        }
    }
}

/// <devdoc>
///     <para>Gets the collection of custom user information.</para>
/// </devdoc>
[
ResCategoryAttribute(Res.DataCategory_Data),
Browsable(false),
ResDescriptionAttribute(Res.ExtendedPropertiesDescr)
]
public PropertyCollection ExtendedProperties {
    get {
        if (extendedProperties == null) {
            extendedProperties = new PropertyCollection();
        }
        return extendedProperties;
    }
}

/// <devdoc>
///     <para>
///         Gets a value indicating whether there are errors in any
///         of the rows in any of the tables of this <see

```

```

    cref='System.Data.DataSet' /> .
    ///     </para>
    /// </devdoc>
    [Browsable(false), ResDescriptionAttribute(Res.DataSetHasErrorsDescr)]
    public bool HasErrors {
        get {
            for (int i = 0; i < Tables.Count; i++) {
                if (Tables[i].HasErrors)
                    return true;
            }
            return false;
        }
    }

    [Browsable(false)]
    public bool IsInitialized {
        get {
            return !fInitInProgress;
        }
    }

    /// <devdoc>
    ///     <para>
    ///         Gets or sets the locale information used to compare strings within the
table.
    ///     </para>
    /// </devdoc>
    [
        ResCategoryAttribute(Res.DataCategory_Data),
        ResDescriptionAttribute(Res.DataSetLocaleDescr)
    ]
    public CultureInfo Locale {
        get {
            // used for comparing not formating/parsing
            Debug.Assert(null != _culture, "DataSet.Locale: null culture");
            return _culture;
        }
        set {
            IntPtr hscp;
            Bid.ScopeEnter(out hscp, "<ds.DataSet.set_Locale|API> %d#\n", ObjectID);
            try {
                if (value != null) {
                    if (!_culture.Equals(value)) {
                        SetLocaleValue(value, true);
                    }
                }
            }
            finally {
                Bid.ScopeExit(hscp);
            }
        }
    }

```

```

        }
        _cultureUserSet = true;
    }
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}
}

```

```

internal void SetLocaleValue(CultureInfo value, bool userSet) {

```

```

    bool flag = false;

```

```

    bool exceptionThrown = false;

```

```

    int tableCount = 0;

```

```

    CultureInfo oldLocale = _culture;

```

```

    bool oldUserSet = _cultureUserSet;

```

```

    try {

```

```

        _culture = value;

```

```

        _cultureUserSet = userSet;

```

```

        foreach (DataTable table in Tables) {

```

```

            if (!table.ShouldSerializeLocale()) {

```

```

                bool retchk = table.SetLocaleValue(value, false, false);

```

```

                //Debug.Assert(retchk == table.ShouldSerializeLocale(),

```

```

                "unexpected setting of table locale"); may fire with Deserialize

```

```

            }

```

```

        }

```

```

        flag = ValidateLocaleConstraint();

```

```

        if (flag) {

```

```

            flag = false;

```

```

            foreach (DataTable table in Tables) {

```

```

                tableCount++;

```

```

                if (!table.ShouldSerializeLocale()) {

```

```

                    table.SetLocaleValue(value, false, true);

```

```

                }

```

```

            }

```

```

            flag = true;

```

```

        }

```

```

    }

```

```

    catch {

```

```

        exceptionThrown = true;

```

```

        throw;
    }
    finally {
        if (!flag) { // reset old locale if ValidationFailed or exception thrown
            _culture = oldLocale;
            _cultureUserSet = oldUserSet;
            foreach (DataTable table in Tables) {
                if (!table.ShouldSerializeLocale()) {
                    table.SetLocaleValue(oldLocale, false, false);
                }
            }
            try {
                for (int i = 0; i < tableCount; ++i) {
                    if (!Tables[i].ShouldSerializeLocale()) {
                        Tables[i].SetLocaleValue(oldLocale, false, true);
                    }
                }
            }
            catch (Exception e) {
                if (!Common.ADP.IsCatchableExceptionType(e)) {
                    throw;
                }
                Common.ADP.TraceExceptionWithoutRethrow(e);
            }
            if (!exceptionThrown) {
                throw ExceptionBuilder.CannotChangeCaseLocale(null);
            }
        }
    }
}

```

```

internal bool ShouldSerializeLocale() {
    // this method is used design-time scenarios via reflection
    // by the property grid to show the Locale property in bold or not
    // by the code dom for persisting the Locale property or not

    // we always want the locale persisted if set by user or different the
current thread
    // but that logic should be performed by the serialization code
    return _cultureUserSet;
}

```

```

/// <devdoc>
///     <para>[To be supplied.]</para>

```

```

    /// </devdoc>
    [Browsable(false),
    DesignerSerializationVisibility(DesignerSerializationVisibility.Hidden)]
    public override ISite Site {
        get {
            return base.Site;
        }
        set {
            ISite oldSite = Site;
            if (value == null && oldSite != null) {
                IContainer cont = oldSite.Container;

                if (cont != null) {
                    for (int i = 0; i < Tables.Count; i++) {
                        if (Tables[i].Site != null) {
                            cont.Remove(Tables[i]);
                        }
                    }
                }
            }
            base.Site = value;
        }
    }

    /// <devdoc>
    ///     <para>
    ///         Get the collection of relations that link tables and
    ///         allow navigation from parent tables to child tables.
    ///     </para>
    /// </devdoc>
    [
    DesignerSerializationVisibility(DesignerSerializationVisibility.Content),
    ResCategoryAttribute(Res.DataCategory_Data),
    ResDescriptionAttribute(Res.DataSetRelationsDescr)
    ]
    public DataRelationCollection Relations {
        get {
            return relationCollection;
        }
    }
}

    /// <devdoc>

```

```

    /// <para>
    ///     Indicates whether <see cref='Relations'> property should be persisted.
    /// </para>
    /// </devdoc>
    protected virtual bool ShouldSerializeRelations() {
        return true; /*Relations.Count > 0;*/ // VS7 300569
    }

    /// <devdoc>
    /// <para>
    ///     Resets the <see cref='System.Data.DataSet.Relations'> property to its
default state.
    /// </para>
    /// </devdoc>
    private void ResetRelations()
    {
        Relations.Clear();
    }

    /// <devdoc>
    /// <para>Gets the collection of tables contained in the <see
cref='System.Data.DataSet'>.</para>
    /// </devdoc>
    [
        DesignerSerializationVisibility(DesignerSerializationVisibility.Content),
        ResCategoryAttribute(Res.DataCategory_Data),
        ResDescriptionAttribute(Res.DataSetTablesDescr)
    ]
    public DataTableCollection Tables {
        get {
            return tableCollection;
        }
    }

    /// <devdoc>
    /// <para>
    ///     Indicates whether <see cref='System.Data.DataSet.Tables'> property
should be persisted.
    /// </para>
    /// </devdoc>
    protected virtual bool ShouldSerializeTables() {
        return true; /*(Tables.Count > 0);*/ // VS7 300569
    }

```

```

    /// <devdoc>
    ///     <para>
    ///         Resets the <see cref='System.Data.DataSet.Tables'> property to its
default state.
    ///     </para>
    /// </devdoc>
    private void ResetTables()
    {
        Tables.Clear();
    }

    internal bool FBoundToDocument {
        get {
            return fBoundToDocument;
        }
        set {
            fBoundToDocument = value;
        }
    }

    /// <devdoc>
    ///     <para>
    ///         Commits all the changes made to this <see cref='System.Data.DataSet'>
since it was loaded or the last
    ///         time <see cref='System.Data.DataSet.AcceptChanges'> was called.
    ///     </para>
    /// </devdoc>
    public void AcceptChanges()
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.AcceptChanges|API> %d#\n", ObjectID);
        try {
            for (int i = 0; i < Tables.Count; i++)
                Tables[i].AcceptChanges();
        }
        finally {
            Bid.ScopeLeave(ref hscp);
        }
    }

    internal event PropertyChangedEventHandler PropertyChanging {
        add {
            onPropertyChangingDelegate += value;
        }
    }

```

```

        remove {
            onPropertyChangingDelegate -= value;
        }
    }

    /// <devdoc>
    ///     <para>Occurs when attempting to merge schemas for two tables with the same
    ///         name.</para>
    /// </devdoc>
    [
        ResCategoryAttribute(Res.DataCategory_Action),
        ResDescriptionAttribute(Res.DataSetMergeFailedDescr)
    ]
    public event MergeFailedEventHandler MergeFailed {
        add {
            onMergeFailed += value;
        }
        remove {
            onMergeFailed -= value;
        }
    }

    internal event DataRowCreatedEventHandler DataRowCreated {
        add {
            onDataRowCreated += value;
        }
        remove {
            onDataRowCreated -= value;
        }
    }

    internal event DataSetClearEventHandler ClearFunctionCalled {
        add {
            onClearFunctionCalled += value;
        }
        remove {
            onClearFunctionCalled -= value;
        }
    }

    [
        ResCategoryAttribute(Res.DataCategory_Action),
        ResDescriptionAttribute(Res.DataSetInitializedDescr)
    ]

```

```

public event System.EventHandler Initialized {
    add {
        onInitialized += value;
    }
    remove {
        onInitialized -= value;
    }
}

public void BeginInit() {
    fInitInProgress = true;
}

public void EndInit() {
    Tables.FinishInitCollection();
    for (int i = 0; i < Tables.Count; i++) {
        Tables[i].Columns.FinishInitCollection();
    }

    for (int i = 0; i < Tables.Count; i++) {
        Tables[i].Constraints.FinishInitConstraints();
    }
}

```

```

((DataRelationCollection.DataSetRelationCollection)Relations).FinishInitRelations();

```

```

    fInitInProgress = false;
    OnInitialized();
}

```

```

/// <devdoc>

```

```

/// <para>Clears the <see cref='System.Data.DataSet'/> of any data by removing
all rows in all tables.</para>

```

```

/// </devdoc>

```

```

public void Clear()

```

```

{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.Clear|API> %d#\n", ObjectID);
    try {
        OnClearFunctionCalled(null);
        bool fEnforce = EnforceConstraints;
        EnforceConstraints = false;
        for (int i = 0; i < Tables.Count; i++)
            Tables[i].Clear();
    }
}

```

```

        EnforceConstraints = fEnforce;
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

/// <devdoc>
/// <para>Clones the structure of the <see cref='System.Data.DataSet'/>,
including all <see cref='System.Data.DataTable'/> schemas, relations, and
///     constraints.</para>
/// </devdoc>
/// Prevent inlining so that reflection calls are not moved to caller that may be
in a different assembly that may have a different grant set.
[MethodImpl(MethodImplOptions.NoInlining)]
public virtual DataSet Clone() {
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.Clone|API> %d#\n", ObjectID);
    try {
        DataSet ds = (DataSet)Activator.CreateInstance(this.GetType(), true);

        if (ds.Tables.Count > 0) // Microsoft : To clean up all the schema in
strong typed dataset.
            ds.Reset();

        //copy some original dataset properties
        ds.DataSetName = DataSetName;
        ds.CaseSensitive = CaseSensitive;
        ds._culture = _culture;
        ds._cultureUserSet = _cultureUserSet;
        ds.EnforceConstraints = EnforceConstraints;
        ds.Namespace = Namespace;
        ds.Prefix = Prefix;
        ds.RemotingFormat = RemotingFormat;
        ds.fIsSchemaLoading = true; //delay expression evaluation

        // ...Tables...
        DataTableCollection tbls = Tables;
        for (int i = 0; i < tbls.Count; i++) {
            DataTable dt = tbls[i].Clone(ds);
            dt.tableNamespace = tbls[i].Namespace; // hardcode the namespace for
a second to not mess up
            // DataRelation cloning.

```

```

        ds.Tables.Add(dt);
    }

    // ...Constraints...
    for (int i = 0; i < tbls.Count; i++) {
        ConstraintCollection constraints = tbls[i].Constraints;
        for (int j = 0; j < constraints.Count; j++) {
            if (constraints[j] is UniqueConstraint)
                continue;

            ForeignKeyConstraint foreign = constraints[j] as
ForeignKeyConstraint;
            if (foreign.Table == foreign.RelatedTable)
                continue; // we have already added this foreign key in while
cloning the datatable

            ds.Tables[i].Constraints.Add(constraints[j].Clone(ds));
        }
    }

    // ...Relations...
    DataRelationCollection rels = Relations;
    for (int i = 0; i < rels.Count; i++) {
        DataRelation rel = rels[i].Clone(ds);
        rel.CheckMultipleNested = false; // disable the check for multiple
nested parent

        ds.Relations.Add(rel);
        rel.CheckMultipleNested = true; // enable the check for multiple
nested parent
    }

    // ...Extended Properties...
    if (this.extendedProperties != null) {
        foreach (Object key in this.extendedProperties.Keys) {
            ds.ExtendedProperties[key] = this.extendedProperties[key];
        }
    }

    foreach (DataTable table in Tables)
        foreach (DataColumn col in table.Columns)
            if (col.Expression.Length != 0)
                ds.Tables[table.TableName,
table.Namespace].Columns[col.ColumnName].Expression = col.Expression;

    for (int i = 0; i < tbls.Count; i++) {
        ds.Tables[i].tableNamespace = tbls[i].tableNamespace; // undo the

```

hardcoding of the namespace

```
    }

    ds.fIsSchemaLoading = false; //reactivate column computations

    return ds;
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

/// <devdoc>
/// <para>Copies both the structure and data for this &ltsee
cref='System.Data.DataSet' />.</para>
/// </devdoc>
public DataSet Copy()
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.Copy|API> %d#\n", ObjectID);
    try {
        DataSet dsNew = Clone();
        bool fEnforceConstraints = dsNew.EnforceConstraints;
        dsNew.EnforceConstraints = false;
        foreach (DataTable table in this.Tables)
        {
            DataTable destTable = dsNew.Tables[table.TableName, table.Namespace];

            foreach (DataRow row in table.Rows)
                table.CopyRow(destTable, row);
        }

        dsNew.EnforceConstraints = fEnforceConstraints;

        return dsNew;
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

internal Int32 EstimatedXmlStringSize()
{
    Int32 bytes = 100;
```

```

        for (int i = 0; i < Tables.Count; i++) {
            Int32 rowBytes = (Tables[i].TableName.Length + 4) << 2;
            DataTable table = Tables[i];
            for (int j = 0; j < table.Columns.Count; j++) {
                rowBytes += ((table.Columns[j].ColumnName.Length + 4) << 2);
                rowBytes += 20;
            }
            bytes += table.Rows.Count * rowBytes;
        }

        return bytes;
    }

    /// <devdoc>
    /// <para>Returns a copy of the <see cref='System.Data.DataSet'> that contains
all changes made to
    ///     it since it was loaded or <see cref='System.Data.DataSet.AcceptChanges'>
    ///     was last called.</para>
    /// </devdoc>
    public DataSet GetChanges()
    {
        return GetChanges(DataRowState.Added | DataRowState.Deleted |
DataRowState.Modified);
    }

    private struct TableChanges {
        private BitArray _rowChanges;
        private int _hasChanges;

        internal TableChanges(int rowCount) {
            _rowChanges = new BitArray(rowCount);
            _hasChanges = 0;
        }

        internal int HasChanges {
            get { return _hasChanges; }
            set { _hasChanges = value; }
        }

        internal bool this[int index] {
            get { return _rowChanges[index]; }
            set {
                Debug.Assert(value && !_rowChanges[index], "setting twice or to
false");
                _rowChanges[index] = value;
                _hasChanges++;
            }
        }
    }

```

```

    }
}

public DataSet GetChanges(DataRowState rowStates)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.GetChanges|API> %d#",
rowStates=%d{ds.DataRowState}\n", ObjectID, (int)rowStates);
    try {
        DataSet dsNew = null;
        bool fEnforceConstraints = false;
        if (0 != (rowStates & ~(DataRowState.Added | DataRowState.Deleted |
DataRowState.Modified | DataRowState.Unchanged))) {
            throw ExceptionBuilder.InvalidRowState(rowStates);
        }

        // Initialize all the individual table bitmaps.
        TableChanges[] bitMatrix = new TableChanges[Tables.Count];
        for (int i = 0; i < bitMatrix.Length; ++i) {
            bitMatrix[i] = new TableChanges(Tables[i].Rows.Count);
        }

        // find all the modified rows and their parents
        MarkModifiedRows(bitMatrix, rowStates);

        // copy the changes to a cloned table
        for (int i = 0; i < bitMatrix.Length; ++i) {
            Debug.Assert(0 <= bitMatrix[i].HasChanges, "negative change count");
            if (0 < bitMatrix[i].HasChanges) {
                if (null == dsNew) {
                    dsNew = this.Clone();
                    fEnforceConstraints = dsNew.EnforceConstraints;
                    dsNew.EnforceConstraints = false;
                }

                DataTable table = this.Tables[i];
                DataTable destTable = dsNew.Tables[table.TableName,
table.Namespace];

                Debug.Assert(bitMatrix[i].HasChanges <= table.Rows.Count, "to
many changes");

                for (int j = 0; 0 < bitMatrix[i].HasChanges; ++j) { // Loop
through the rows.

```

```

        if (bitMatrix[i][j]) {
            table.CopyRow(destTable, table.Rows[j]);
            bitMatrix[i].HasChanges--;
        }
    }
}

if (null != dsNew) {
    dsNew.EnforceConstraints = fEnforceConstraints;
}
return dsNew;
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

```

```

private void MarkModifiedRows(TableChanges[] bitMatrix, DataRowState rowStates) {
    // for every table, every row & every relation find the modified rows and for
non-deleted rows, their parents
    for (int tableIndex = 0; tableIndex < bitMatrix.Length; ++tableIndex) {
        DataRowCollection rows = Tables[tableIndex].Rows;
        int rowCount = rows.Count;

        for (int rowIndex = 0; rowIndex < rowCount; ++rowIndex) {
            DataRow row = rows[rowIndex];
            DataRowState rowState = row.RowState;
            Debug.Assert(DataRowState.Added == rowState ||
                DataRowState.Deleted == rowState ||
                DataRowState.Modified == rowState ||
                DataRowState.Unchanged == rowState,
                "unexpected DataRowState");

            // if bit not already set and row is modified
            if ((0 != (rowStates & rowState)) && !bitMatrix[tableIndex]
[roIndex]) {

                bitMatrix[tableIndex][rowIndex] = true;

                if (DataRowState.Deleted != rowState) {
                    MarkRelatedRowsAsModified(bitMatrix, row);
                }
            }
        }
    }
}

```

```

    }
}

private void MarkRelatedRowsAsModified(TableChanges[] bitMatrix, DataRow row) {
    DataRelationCollection relations = row.Table.ParentRelations;
    int relationCount = relations.Count;
    for (int relatedIndex = 0; relatedIndex < relationCount; ++relatedIndex) {
        DataRow[] relatedRows = row.GetParentRows(relations[relatedIndex],
DataRowVersion.Current);

        foreach (DataRow relatedRow in relatedRows) {
            int relatedTableIndex = this.Tables.IndexOf(relatedRow.Table);
            int relatedRowIndex = relatedRow.Table.Rows.IndexOf(relatedRow);

            if (!bitMatrix[relatedTableIndex][relatedRowIndex]) {
                bitMatrix[relatedTableIndex][relatedRowIndex] = true;

                if (DataRowState.Deleted != relatedRow.RowState) {
                    // recurse into related rows
                    MarkRelatedRowsAsModified(bitMatrix, relatedRow);
                }
            }
        }
    }
}

```

```

IList System.ComponentModel.IListSource.GetList() {
    return DefaultViewManager;
}

```

```

internal string GetRemotingDiffGram(DataTable table)
{
    StringWriter strWriter = new StringWriter(CultureInfo.InvariantCulture);
    XmlTextWriter writer = new XmlTextWriter(strWriter);
    writer.Formatting = Formatting.Indented;
    if (strWriter != null) {
        // Create and save the updates
        new NewDiffgramGen(table, false).Save(writer, table);
    }

    return strWriter.ToString();
}

```

```

public string GetXml()

```

```

{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.GetXml|API> %d#\n", ObjectID);
    try {

        // StringBuilder strBuilder = new
StringBuilder(EstimatedXmlStringSize());
        // StringWriter strWriter = new StringWriter(strBuilder);
        StringWriter strWriter = new StringWriter(CultureInfo.InvariantCulture);
        if (strWriter != null) {
            XmlTextWriter w = new XmlTextWriter(strWriter);
            w.Formatting = Formatting.Indented;
            new XmlDataTreeWriter(this).Save(w, false);
        }
        return strWriter.ToString();
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

public string GetXmlSchema()
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.GetXmlSchema|API> %d#\n", ObjectID);
    try {
        StringWriter strWriter = new StringWriter(CultureInfo.InvariantCulture);
        XmlTextWriter writer = new XmlTextWriter(strWriter);
        writer.Formatting = Formatting.Indented;
        if (strWriter != null) {
            (new XmlTreeGen(SchemaFormat.Public)).Save(this, writer);
        }

        return strWriter.ToString();
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

internal string GetXmlSchemaForRemoting(DataTable table)
{
    StringWriter strWriter = new StringWriter(CultureInfo.InvariantCulture);
    XmlTextWriter writer = new XmlTextWriter(strWriter);

```

```

        writer.Formatting = Formatting.Indented;
        if (strWriter != null) {
            if (table == null) {
                if (this.SchemaSerializationMode ==
SchemaSerializationMode.ExcludeSchema)
                    (new XmlTreeGen(SchemaFormat.RemotingSkipSchema)).Save(this,
writer);

                else
                    (new XmlTreeGen(SchemaFormat.Remoting)).Save(this, writer);
            }
            else { // no skip schema support for typed datatable
                (new XmlTreeGen(SchemaFormat.Remoting)).Save(table, writer);
            }
        }

        return strWriter.ToString();
    }

    /// <devdoc>
    /// <para>Gets a value indicating whether the <see cref='System.Data.DataSet'>
has changes, including new,
    ///     deleted, or modified rows.</para>
    /// </devdoc>
    public bool HasChanges()
    {
        return HasChanges(DataRowState.Added | DataRowState.Deleted |
DataRowState.Modified);
    }

    /// <devdoc>
    /// <para>Gets a value indicating whether the <see cref='System.Data.DataSet'>
has changes, including new,
    ///     deleted, or modified rows, filtered by <see
cref='System.Data.DataRowState'>.</para>
    /// </devdoc>
    public bool HasChanges(DataRowState rowStates)
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.HasChanges|API> %d#",
rowStates=%d{ds.DataRowState}\n", ObjectID, (int)rowStates);

        try {
            const DataRowState allRowStates = DataRowState.Detached |

```

```
DataRowState.Unchanged | DataRowState.Added | DataRowState.Deleted |  
DataRowState.Modified;
```

```
        if ((rowStates & (~allRowStates)) != 0) {  
            throw ExceptionBuilder.ArgumentOutOfRange("rowState");  
        }  
  
        for (int i = 0; i < Tables.Count; i++) {  
            DataTable table = Tables[i];  
  
            for (int j = 0; j < table.Rows.Count; j++) {  
                DataRow row = table.Rows[j];  
                if ((row.RowState & rowStates) != 0) {  
                    return true;  
                }  
            }  
        }  
        return false;  
    }  
    finally {  
        Bid.ScopeLeave(ref hscp);  
    }  
}  
  
/// <devdoc>  
/// <para>Infer the XML schema from the specified <see  
cref='System.IO.TextReader' /> into the <see cref='System.Data.DataSet' />.</para>  
/// </devdoc>  
public void InferXmlSchema(XmlReader reader, string[] nsArray)  
{  
    IntPtr hscp;  
    Bid.ScopeEnter(out hscp, "<ds.DataSet.InferXmlSchema|API> %d#\n", ObjectID);  
    try {  
        if (reader == null)  
            return;  
  
        XmlDocument xdoc = new XmlDocument();  
        if (reader.NodeType == XmlNodeType.Element) {  
            XmlNode node = xdoc.ReadNode(reader);  
            xdoc.AppendChild(node);  
        }  
        else  
            xdoc.Load(reader);  
        if (xdoc.DocumentElement == null)
```

```

        return;

        InferSchema(xdoc, nsArray, XmlReadMode.InferSchema);
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

/// <devdoc>
/// <para>Infer the XML schema from the specified <see
 cref='System.IO.TextReader' /> into the <see cref='System.Data.DataSet' />.</para>
/// </devdoc>
public void InferXmlSchema(Stream stream, string[] nsArray)
{
    if (stream == null)
        return;

    InferXmlSchema(new XmlTextReader(stream), nsArray);
}

/// <devdoc>
/// <para>Infer the XML schema from the specified <see
 cref='System.IO.TextReader' /> into the <see cref='System.Data.DataSet' />.</para>
/// </devdoc>
public void InferXmlSchema(TextReader reader, string[] nsArray)
{
    if (reader == null)
        return;

    InferXmlSchema(new XmlTextReader(reader), nsArray);
}

/// <devdoc>
/// <para>Infer the XML schema from the specified file into the <see
 cref='System.Data.DataSet' />.</para>
/// </devdoc>
[ResourceExposure(ResourceScope.Machine)]
[ResourceConsumption(ResourceScope.Machine)]
public void InferXmlSchema(String fileName, string[] nsArray)
{
    XmlTextReader xr = new XmlTextReader(fileName);
    try {
        InferXmlSchema(xr, nsArray);
    }
}

```

```

    }
    finally {
        xr.Close();
    }
}

/// <devdoc>
/// <para>Reads the XML schema from the specified <see
cref='T:System.Xml.XmlReader' /> into the <see cref='System.Data.DataSet' />
/// .</para>
/// </devdoc>
public void ReadXmlSchema(XmlReader reader)
{
    ReadXmlSchema(reader, false);
}

internal void ReadXmlSchema(XmlReader reader, bool denyResolving)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.ReadXmlSchema|INFO> %d#, reader,
denyResolving=%d{bool}\\n", ObjectID, denyResolving);
    try {
        int iCurrentDepth = -1;

        if (reader == null)
            return;

        if (reader is XmlTextReader)
            ((XmlTextReader)reader).WhitespaceHandling = WhitespaceHandling.None;

        XmlDocument xdoc = new XmlDocument(); // we may need this to infer the
schema

        if (reader.NodeType == XmlNodeType.Element)
            iCurrentDepth = reader.Depth;

        reader.MoveToContent();

        if (reader.NodeType == XmlNodeType.Element) {
            // if reader points to the schema load it...

            if (reader.LocalName == Keywords.XDR_SCHEMA && reader.NamespaceURI ==
Keywords.XDRNS) {
                // load XDR schema and exit

```

```

        ReadXDRSchema(reader);
        return;
    }

    if (reader.LocalName == Keywords.XSD_SCHEMA && reader.NamespaceURI ==
Keywords.XSDNS) {
        // load XSD schema and exit
        ReadXSDSchema(reader, denyResolving);
        return;
    }

    if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI.StartsWith(Keywords.XSD_NS_START, StringComparison.Ordinal))
        throw ExceptionBuilder.DataSetUnsupportedSchema(Keywords.XSDNS);

    // ... otherwise backup the top node and all its attributes
    XmlElement topNode = xdoc.CreateElement(reader.Prefix,
reader.LocalName, reader.NamespaceURI);
    if (reader.HasAttributes) {
        int attrCount = reader.AttributeCount;
        for (int i = 0; i < attrCount; i++) {
            reader.MoveToAttribute(i);
            if (reader.NamespaceURI.Equals(Keywords.XSD_XMLNS_NS))
                topNode.SetAttribute(reader.Name,
reader.GetAttribute(i));
            else {
                XmlAttribute attr =
topNode.SetAttributeNode(reader.LocalName, reader.NamespaceURI);
                attr.Prefix = reader.Prefix;
                attr.Value = reader.GetAttribute(i);
            }
        }
    }
    reader.Read();

    while (MoveToElement(reader, iCurrentDepth)) {

        // if reader points to the schema load it...
        if (reader.LocalName == Keywords.XDR_SCHEMA &&
reader.NamespaceURI == Keywords.XDRNS) {
            // load XDR schema and exit
            ReadXDRSchema(reader);
            return;
        }
    }

```

```

        if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI == Keywords.XSDNS) {
            // load XSD schema and exit
            ReadXSDSchema(reader, denyResolving);
            return;
        }

        if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI.StartsWith(Keywords.XSD_NS_START, StringComparison.Ordinal))
            throw
ExceptionBuilder.DataSetUnsupportedSchema(Keywords.XSDNS);

        XmlNode node = xdoc.ReadNode(reader);
        topNode.AppendChild(node);

    }

    // read the closing tag of the current element
    ReadEndElement(reader);

    // if we are here no schema has been found
    xdoc.AppendChild(topNode);

    // so we InferSchema
    InferSchema(xdoc, null, XmlReadMode.Auto);
}
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

internal bool MoveToElement(XmlReader reader, int depth) {
    while (!reader.EOF && reader.NodeType != XmlNodeType.EndElement &&
reader.NodeType != XmlNodeType.Element && reader.Depth > depth) {
        reader.Read();
    }
    return (reader.NodeType == XmlNodeType.Element);
}

private static void MoveToElement(XmlReader reader) {
    while (!reader.EOF && reader.NodeType != XmlNodeType.EndElement &&

```

```

reader.NodeType != XmlNodeType.Element) {
    reader.Read();
}
}

internal void ReadEndElement(XmlReader reader) {
    while (reader.NodeType == XmlNodeType.Whitespace) {
        reader.Skip();
    }
    if (reader.NodeType == XmlNodeType.None) {
        reader.Skip();
    }
    else if (reader.NodeType == XmlNodeType.EndElement) {
        reader.ReadEndElement();
    }
}

internal void ReadXSDSchema(XmlReader reader, bool denyResolving) {
    XmlSchemaSet sSet = new XmlSchemaSet();

    int schemaFragmentCount = 1;
    //read from current schmema element
    if (reader.LocalName == Keywords.XSD_SCHEMA && reader.NamespaceURI ==
Keywords.XSDNS) {
        if (reader.HasAttributes) {
            string attribValue = reader.GetAttribute(Keywords.MSD_FRAGMENTCOUNT,
Keywords.MSDNS); // this must not move the position
            if (!Common.ADP.IsEmpty(attribValue)) {
                schemaFragmentCount = int.Parse(attribValue, null);
            }
        }
    }

    while (reader.LocalName == Keywords.XSD_SCHEMA && reader.NamespaceURI ==
Keywords.XSDNS) {
        XmlSchema s = XmlSchema.Read(reader, null);
        sSet.Add(s);
        //read the end tag
        ReadEndElement(reader);

        if (--schemaFragmentCount > 0) {
            MoveToElement(reader);
        }
        while (reader.NodeType == XmlNodeType.Whitespace) {
            reader.Skip();

```

```

    }
}
sSet.Compile();
XSDSchema schema = new XSDSchema();
schema.LoadSchema(sSet, this);
}

internal void ReadXDRSchema(XmlReader reader) {
    XmlDocument xdoc = new XmlDocument(); // we may need this to infer the schema
    XmlNode schNode = xdoc.ReadNode(reader);
    xdoc.AppendChild(schNode);
    XDRSchema schema = new XDRSchema(this, false);
    this.DataSetName = xdoc.DocumentElement.LocalName;
    schema.LoadSchema((XmlElement)schNode, this);
}

/// <devdoc>
/// <para>Reads the XML schema from the specified <see cref='System.IO.Stream'>
into the
/// <see cref='System.Data.DataSet'>.</para>
/// </devdoc>
public void ReadXmlSchema(Stream stream)
{
    if (stream == null)
        return;

    ReadXmlSchema(new XmlTextReader(stream), false);
}

/// <devdoc>
/// <para>Reads the XML schema from the specified <see
cref='System.IO.TextReader'> into the <see cref='System.Data.DataSet'>.</para>
/// </devdoc>
public void ReadXmlSchema(TextReader reader)
{
    if (reader == null)
        return;

    ReadXmlSchema(new XmlTextReader(reader), false);
}

/// <devdoc>
/// <para>Reads the XML schema from the specified file into the <see
cref='System.Data.DataSet'>.</para>

```

```

    /// </devdoc>
    [ResourceExposure(ResourceScope.Machine)]
    [ResourceConsumption(ResourceScope.Machine)]
    public void ReadXmlSchema(String fileName)
    {
        XmlTextReader xr = new XmlTextReader(fileName);
        try {
            ReadXmlSchema(xr, false);
        }
        finally {
            xr.Close();
        }
    }

    #region WriteXmlSchema
    /// <summary>Writes the <see cref='DataSet'/> structure as an XML schema to using
    the specified <see cref='Stream'/> object.</summary>
    /// <param name="stream">A <see cref='Stream'/> object used to write to a file.
    </param>
    public void WriteXmlSchema(Stream stream)
    {
        this.WriteXmlSchema(stream, SchemaFormat.Public, (Converter<Type,
string>)null);
    }

    /// <summary>Writes the <see cref='DataSet'/> structure as an XML schema to using
    the specified <see cref='Stream'/> object.</summary>
    /// <param name="stream">A <see cref='Stream'/> object used to write to a file.
    </param>
    /// <param name="multipleTargetConverter">A delegate used to convert <see
    cref='Type'/> into string.</param>
    public void WriteXmlSchema(Stream stream, Converter<Type, string>
multipleTargetConverter)
    {
        ADP.CheckArgumentNull(multipleTargetConverter, "multipleTargetConverter");
        this.WriteXmlSchema(stream, SchemaFormat.Public, multipleTargetConverter);
    }

    /// <summary>Writes the <see cref='DataSet'/> structure as an XML schema to a
    file.</summary>
    /// <param name="fileName">The file name (including the path) to which to write.
    </param>
    [ResourceExposure(ResourceScope.Machine)]
    [ResourceConsumption(ResourceScope.Machine)]

```

```

    public void WriteXmlSchema(String fileName)
    {
        this.WriteXmlSchema(fileName, SchemaFormat.Public, (Converter<Type,
string>)null);
    }

    /// <summary>Writes the <see cref='DataSet'> structure as an XML schema to a
file.</summary>
    /// <param name="fileName">The file name (including the path) to which to write.
</param>
    /// <param name="multipleTargetConverter">A delegate used to convert <see
cref='Type'> into string.</param>
    [ResourceExposure(ResourceScope.Machine)]
    [ResourceConsumption(ResourceScope.Machine)]
    public void WriteXmlSchema(String fileName, Converter<Type, string>
multipleTargetConverter)
    {
        ADP.CheckArgumentNull(multipleTargetConverter, "multipleTargetConverter");
        this.WriteXmlSchema(fileName, SchemaFormat.Public, multipleTargetConverter);
    }

    /// <summary>Writes the <see cref='DataSet'> structure as an XML schema to a
<see cref='TextWriter'> object.</summary>
    /// <param name="writer">The <see cref='TextWriter'> object with which to write.
</param>
    public void WriteXmlSchema(TextWriter writer)
    {
        this.WriteXmlSchema(writer, SchemaFormat.Public, (Converter<Type,
string>)null);
    }

    /// <summary>Writes the <see cref='DataSet'> structure as an XML schema to a
<see cref='TextWriter'> object.</summary>
    /// <param name="writer">The <see cref='TextWriter'> object with which to write.
</param>
    /// <param name="multipleTargetConverter">A delegate used to convert <see
cref='Type'> into string.</param>
    public void WriteXmlSchema(TextWriter writer, Converter<Type, string>
multipleTargetConverter)
    {
        ADP.CheckArgumentNull(multipleTargetConverter, "multipleTargetConverter");
        this.WriteXmlSchema(writer, SchemaFormat.Public, multipleTargetConverter);
    }

```

```

    /// <summary>Writes the <see cref='DataSet'/> structure as an XML schema to an
<see cref='XmlWriter'/> object.</summary>
    /// <param name="writer">The <see cref='XmlWriter'/> object with which to write.
</param>
    public void WriteXmlSchema(XmlWriter writer)
    {
        this.WriteXmlSchema(writer, SchemaFormat.Public, (Converter<Type,
string>)null);
    }

    /// <summary>Writes the <see cref='DataSet'/> structure as an XML schema to an
<see cref='XmlWriter'/> object.</summary>
    /// <param name="writer">The <see cref='XmlWriter'/> object with which to write.
</param>
    /// <param name="multipleTargetConverter">A delegate used to convert <see
cref='Type'/> into string.</param>
    public void WriteXmlSchema(XmlWriter writer, Converter<Type, string>
multipleTargetConverter)
    {
        ADP.CheckArgumentNull(multipleTargetConverter, "multipleTargetConverter");
        this.WriteXmlSchema(writer, SchemaFormat.Public, multipleTargetConverter);
    }

    [ResourceExposure(ResourceScope.Machine)]
    [ResourceConsumption(ResourceScope.Machine)]
    private void WriteXmlSchema(String fileName, SchemaFormat schemaFormat,
Converter<Type, string> multipleTargetConverter)
    {
        XmlTextWriter xw = new XmlTextWriter( fileName, null );
        try {
            xw.Formatting = Formatting.Indented;
            xw.WriteStartDocument(true);
            this.WriteXmlSchema(xw, schemaFormat, multipleTargetConverter);
            xw.WriteEndDocument();
        }
        finally {
            xw.Close();
        }
    }

    private void WriteXmlSchema(Stream stream, SchemaFormat schemaFormat,
Converter<Type, string> multipleTargetConverter)
    {
        if (stream == null)

```

```

        return;

        XmlTextWriter w = new XmlTextWriter(stream, null);
        w.Formatting = Formatting.Indented;

        this.WriteXmlSchema(w, schemaFormat, multipleTargetConverter);
    }

    private void WriteXmlSchema(TextWriter writer, SchemaFormat schemaFormat,
Converter<Type, string> multipleTargetConverter)
    {
        if (writer == null)
            return;

        XmlTextWriter w = new XmlTextWriter(writer);
        w.Formatting = Formatting.Indented;

        this.WriteXmlSchema(w, schemaFormat, multipleTargetConverter);
    }

    private void WriteXmlSchema(XmlWriter writer, SchemaFormat schemaFormat,
Converter<Type, string> multipleTargetConverter)
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.WriteXmlSchema|INFO> %d#,
schemaFormat=%d{ds.SchemaFormat}\n", ObjectID, (int)schemaFormat);
        try {
            // Generate SchemaTree and write it out
            if (writer != null) {
                XmlTreeGen treeGen = null;
                if (schemaFormat == SchemaFormat.WebService &&
SchemaSerializationMode == SchemaSerializationMode.ExcludeSchema
&&
                    writer.WriteState == WriteState.Element) {
                    treeGen = new XmlTreeGen(SchemaFormat.WebServiceSkipSchema);
                }
                else {
                    treeGen = new XmlTreeGen(schemaFormat);
                }

                treeGen.Save(this, (DataTable)null, writer, false,
multipleTargetConverter);
            }
        }
    }

```

```

        finally {
            Bid.ScopeLeave(ref hscp);
        }
    }
#endregion

/// <devdoc>
/// </devdoc>
public XmlReadMode ReadXml(XmlReader reader)
{
    return ReadXml(reader, false);
}

internal XmlReadMode ReadXml(XmlReader reader, bool denyResolving)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.ReadXml|INFO> %d#,
denyResolving=%d{bool}\\n", ObjectID, denyResolving);
    try {

        DataTable.DSRowDiffIdUsageSection rowDiffIdUsage = new
DataTable.DSRowDiffIdUsageSection();
        try {
            bool fDataFound = false;
            bool fSchemaFound = false;
            bool fDiffsFound = false;
            bool fIsXdr = false;
            int iCurrentDepth = -1;
            XmlReadMode ret = XmlReadMode.Auto;
            bool isEmptyDataSet = false;
            bool topNodeIsProcessed = false; // we chance topnode and there is
just one case that we miss to process it
            // it is : <elem attrib1="Attrib">txt</elem>

            // clear the hashtable to avoid conflicts between diffgrams,
SqlHotFix 782

            rowDiffIdUsage.Prepare(this);

            if (reader == null)
                return ret;

            if (Tables.Count == 0) {
                isEmptyDataSet = true;

```

```

    }

    if (reader is XmlTextReader)
        ((XmlTextReader)reader).WhitespaceHandling =
WhitespaceHandling.Significant;

    XmlDocument xdoc = new XmlDocument(); // we may need this to infer
the schema

    XmlDataLoader xmlload = null;

    reader.MoveToContent();

    if (reader.NodeType == XmlNodeType.Element)
        iCurrentDepth = reader.Depth;

    if (reader.NodeType == XmlNodeType.Element) {
        if ((reader.LocalName == Keywords.DIFFGRAM) &&
(reader.NamespaceURI == Keywords.DFFNS)) {
            this.ReadXmlDiffgram(reader);
            // read the closing tag of the current element
            ReadEndElement(reader);
            return XmlReadMode.DiffGram;
        }

        // if reader points to the schema load it
        if (reader.LocalName == Keywords.XDR_SCHEMA &&
reader.NamespaceURI == Keywords.XDRNS) {
            // load XDR schema and exit
            ReadXDRSchema(reader);
            return XmlReadMode.ReadSchema; //since the top level element
is a schema return
        }

        if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI == Keywords.XSDNS) {
            // load XSD schema and exit
            ReadXSDSchema(reader, denyResolving);
            return XmlReadMode.ReadSchema; //since the top level element
is a schema return
        }

        if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI.StartsWith(Keywords.XSD_NS_START, StringComparison.Ordinal))

```

```

        throw
ExceptionBuilder.DataSetUnsupportedSchema(Keywords.XSDNS);

        // now either the top level node is a table and we load it
        through dataReader...

        // ... or backup the top node and all its attributes because we
        may need to InferSchema
        XmlElement topNode = xdoc.CreateElement(reader.Prefix,
reader.LocalName, reader.NamespaceURI);
        if (reader.HasAttributes) {
            int attrCount = reader.AttributeCount;
            for (int i = 0; i < attrCount; i++) {
                reader.MoveToAttribute(i);
                if (reader.NamespaceURI.Equals(Keywords.XSD_XMLNS_NS))
                    topNode.SetAttribute(reader.Name,
reader.GetAttribute(i));
            }
            else {
                XmlAttribute attr =
topNode.SetAttributeNode(reader.LocalName, reader.NamespaceURI);
                attr.Prefix = reader.Prefix;
                attr.Value = reader.GetAttribute(i);
            }
        }
        reader.Read();
        string rootNodeSimpleContent = reader.Value;

        while (MoveToElement(reader, iCurrentDepth)) {

            if ((reader.LocalName == Keywords.DIFFGRAM) &&
(reader.NamespaceURI == Keywords.DFFNS)) {
                this.ReadXmlDiffgram(reader);
                // read the closing tag of the current element
                // YUKON FIX
                ReadEndElement(reader);

                // return
                XmlReadMode.DiffGram;

                ret = XmlReadMode.DiffGram; // continue reading for
multiple schemas
            }

            // if reader points to the schema load it...

```

```

        if (!fSchemaFound && !fDataFound && reader.LocalName ==
Keywords.XDR_SCHEMA && reader.NamespaceURI == Keywords.XDRNS) {
            // load XDR schema and exit
            ReadXDRSchema(reader);
            fSchemaFound = true;
            fIsXdr = true;
            continue;
        }

        if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI == Keywords.XSDNS) {
            // load XSD schema and exit
            ReadXSDSchema(reader, denyResolving);
            fSchemaFound = true;
            continue;
        }

        if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI.StartsWith(Keywords.XSD_NS_START, StringComparison.Ordinal))
            throw
ExceptionBuilder.DataSetUnsupportedSchema(Keywords.XSDNS);

        if ((reader.LocalName == Keywords.DIFFGRAM) &&
(reader.NamespaceURI == Keywords.DFFNS)) {
            this.ReadXmlDiffgram(reader);
            fDiffsFound = true;
            ret = XmlReadMode.DiffGram;
        }
        else {
            // We have found data IFF the reader.NodeType == Element
and reader.depth == currentDepth-1
            // if reader.NodeType == whitespace, skip all white
spaces.

            // skip processing i.e. continue if the first non-
whitespace node is not of type element.
            while (!reader.EOF && reader.NodeType ==
XmlNodeType.Whitespace)

                reader.Read();
            if (reader.NodeType != XmlNodeType.Element)
                continue;
            // we found data here
            fDataFound = true;

```

```

        if (!fSchemaFound && Tables.Count == 0) {
            XmlNode node = xdoc.ReadNode(reader);
            topNode.AppendChild(node);
        }
        else {
            if (xmlload == null)
                xmlload = new XmlDataLoader(this, fIsXdr,
topNode, false);

            xmlload.LoadData(reader);
            topNodeIsProcessed = true; // we process the topnode
            if (fSchemaFound)
                ret = XmlReadMode.ReadSchema;
            else
                ret = XmlReadMode.IgnoreSchema;
        }
    }

    // read the closing tag of the current element
    ReadEndElement(reader);
    bool isfTopLevelTableSet = false;
    bool tmpValue = this.fTopLevelTable;
    //While inference we ignore root elements text content
    if (!fSchemaFound && Tables.Count == 0 && !topNode.HasChildNodes)
    { //We shoule not come add SC of root elemnt to topNode if we are not infering
        this.fTopLevelTable = true;
        isfTopLevelTableSet = true;
        if ((rootNodeSimpleContent != null &&
rootNodeSimpleContent.Length > 0))
            topNode.InnerText = rootNodeSimpleContent;
    }
    if (!isEmptyDataSet) {
        if ((rootNodeSimpleContent != null &&
rootNodeSimpleContent.Length > 0))
            topNode.InnerText = rootNodeSimpleContent;
    }

    // now top node contains the data part
    xdoc.AppendChild(topNode);

    if (xmlload == null)
        xmlload = new XmlDataLoader(this, fIsXdr, topNode, false);

    if (!isEmptyDataSet && !topNodeIsProcessed) {

```

```

XmlElement root = xdoc.DocumentElement;
Debug.Assert(root.NamespaceURI != null, "root.NamespaceURI
should not be null, it should be empty string");
// just recognize that below given Xml represents datatable
in toplevel

//<table attr1="foo" attr2="bar"
table_Text="junk">text</table>
// only allow root element with simple content, if any
if (root.ChildNodes.Count == 0 ||
((root.ChildNodes.Count == 1) &&
root.FirstChild.GetType() == typeof(System.Xml.XmlText))) {
    bool initfTopLevelTable = this.fTopLevelTable;
    // if root element maps to a datatable
    // ds and dt cant have the same name and ns at the same
time, how to write to xml

    if (this.DataSetName != root.Name && this.namespaceURI !=
root.NamespaceURI &&
        Tables.Contains(root.Name, (root.NamespaceURI.Length
== 0) ? null : root.NamespaceURI, false, true)) {
        this.fTopLevelTable = true;
    }
    try {
        xmlload.LoadData(xdoc);
    }
    finally {
        this.fTopLevelTable = initfTopLevelTable; // this is
not for inference, we have schema and we were skipping
        // topnode where it was a datatable, We must restore
the value
    }
}
} // above check and below check are orthogonal
// so we InferSchema
if (!fDiffsFound) {
    // Load Data
    if (!fSchemaFound && Tables.Count == 0) {
        InferSchema(xdoc, null, XmlReadMode.Auto);
        ret = XmlReadMode.InferSchema;
        xmlload.FromInference = true;
    }
    try {
        xmlload.LoadData(xdoc);
    }
    finally {
        xmlload.FromInference = false;
    }
}

```

```

        }
    }

    //We dont need this assignement. Once we set it(where we set
it during inference), it won't be changed
    if (isfTopLevelTableSet)
        this.fTopLevelTable = tmpValue;
    }
}

    return ret;
}
finally {
    rowDiffIdUsage.Cleanup();
}
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

/// <devdoc>
/// </devdoc>
public XmlReadMode ReadXml(Stream stream)
{
    if (stream == null)
        return XmlReadMode.Auto;

    XmlTextReader xr = new XmlTextReader(stream);

    // Prevent Dtd entity in dataset
    xr.XmlResolver = null;

    return ReadXml(xr, false);
}

/// <devdoc>
/// </devdoc>
public XmlReadMode ReadXml(TextReader reader)
{
    if (reader == null)
        return XmlReadMode.Auto;

    XmlTextReader xr = new XmlTextReader(reader);

```

```

        // Prevent Dtd entity in dataset
        xr.XmlResolver = null;

        return ReadXml(xr, false);
    }

    /// <devdoc>
    /// </devdoc>
    [ResourceExposure(ResourceScope.Machine)]
    [ResourceConsumption(ResourceScope.Machine)]
    public XmlReadMode ReadXml(string fileName)
    {
        XmlTextReader xr = new XmlTextReader(fileName);

        // Prevent Dtd entity in dataset
        xr.XmlResolver = null;

        try
        {
            return ReadXml(xr, false);
        }
        finally {
            xr.Close();
        }
    }

    internal void InferSchema(XmlDocument xdoc, string[] excludedNamespaces,
        XmlReadMode mode) {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.InferSchema|INFO> %d#,
mode=%d{ds.XmlReadMode}\n", ObjectID, (int)mode);
        try {
            string ns = xdoc.DocumentElement.NamespaceURI;
            if (null == excludedNamespaces) {
                excludedNamespaces = new string[0];
            }
            XmlNodeReader xnr = new XmlIgnoreNamespaceReader(xdoc,
excludedNamespaces);
            System.Xml.Schema.XmlSchemaInference infer = new
System.Xml.Schema.XmlSchemaInference();

            infer.Occurrence = XmlSchemaInference.InferenceOption.Relaxed;

```

```

        if (mode == XmlReadMode.InferTypedSchema)
            infer.TypeInference = XmlSchemaInference.InferenceOption.Restricted;
        else
            infer.TypeInference = XmlSchemaInference.InferenceOption.Relaxed;

        XmlSchemaSet schemaSet = infer.InferSchema(xnr);
        schemaSet.Compile();

        XSDSchema schema = new XSDSchema();
        schema.FromInference = true;

        try {
            schema.LoadSchema(schemaSet, this);
        }
        finally {
            schema.FromInference = false; // this is always false if you are not
calling from inference
        }
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

private bool IsEmpty() {
    foreach (DataTable table in this.Tables)
        if (table.Rows.Count > 0)
            return false;
    return true;
}

private void ReadXmlDiffgram(XmlReader reader) {
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.ReadXmlDiffgram|INFO> %d#\n",
ObjectID);
    try {
        int d = reader.Depth;
        bool fEnforce = this.EnforceConstraints;
        this.EnforceConstraints = false;
        DataSet newDs;
        bool isEmpty = this.IsEmpty();

        if (isEmpty) {
            newDs = this;

```

```

    }
    else {
        newDs = this.Clone();
        newDs.EnforceConstraints = false;
    }

    foreach (DataTable t in newDs.Tables) {
        t.Rows.nullInList = 0;
    }
    reader.MoveToContent();
    if ((reader.LocalName != Keywords.DIFFGRAM) && (reader.NamespaceURI !=
Keywords.DFFNS))
        return;
    reader.Read();
    if (reader.NodeType == XmlNodeType.Whitespace)
        MoveToElement(reader, reader.Depth - 1 /*iCurrentDepth*/); // skip
over whitespaces.

    newDs.fInLoadDiffgram = true;

    if (reader.Depth > d) {
        if ((reader.NamespaceURI != Keywords.DFFNS) && (reader.NamespaceURI
!= Keywords.MSDNS)) {
            //we should be inside the dataset part
            XmlDocument xdoc = new XmlDocument();
            XmlElement node = xdoc.CreateElement(reader.Prefix,
reader.LocalName, reader.NamespaceURI);
            reader.Read();
            if (reader.NodeType == XmlNodeType.Whitespace) {
                MoveToElement(reader, reader.Depth - 1 /*iCurrentDepth*/); //
skip over whitespaces.
            }
            if (reader.Depth - 1 > d) {
                XmlDataLoader xmlload = new XmlDataLoader(newDs, false, node,
false);

                xmlload.isDiffgram = true; // turn on the special processing
                xmlload.LoadData(reader);
            }
            ReadEndElement(reader);
            if (reader.NodeType == XmlNodeType.Whitespace) {
                MoveToElement(reader, reader.Depth - 1 /*iCurrentDepth*/); //
skip over whitespaces.
            }
        }
    }
}

```

```

        Debug.Assert(reader.NodeType != XmlNodeType.Whitespace, "Should not
be on Whitespace node");

        if (((reader.LocalName == Keywords.SQL_BEFORE) &&
(reader.NamespaceURI == Keywords.DFFNS)) ||
            ((reader.LocalName == Keywords.MSD_ERRORS) &&
(reader.NamespaceURI == Keywords.DFFNS)))

        {
            //this will consume the changes and the errors part
            XMLDiffLoader diffLoader = new XMLDiffLoader();
            diffLoader.LoadDiffGram(newDs, reader);
        }

        // get to the closing diff tag
        while (reader.Depth > d) {
            reader.Read();
        }
        // read the closing tag
        ReadEndElement(reader);
    }

    foreach (DataTable t in newDs.Tables) {
        if (t.Rows.nullInList > 0)
            throw ExceptionBuilder.RowInsertMissing(t.TableName);
    }

    newDs.fInLoadDiffgram = false;

    //terrible performance!
    foreach (DataTable t in newDs.Tables) {
        DataRelation[] nestedParentRelations = t.NestedParentRelations;
        foreach (DataRelation rel in nestedParentRelations) {
            if (rel.ParentTable == t) {
                foreach (DataRow r in t.Rows) {
                    foreach (DataRelation rel2 in nestedParentRelations) {
                        r.CheckForLoops(rel2);
                    }
                }
            }
        }
    }

    if (!isEmpty) {

```

```

        this.Merge(newDs);
        if (this.dataSetName == "NewDataSet")
            this.dataSetName = newDs.dataSetName;
        newDs.EnforceConstraints = fEnforce;
    }
    this.EnforceConstraints = fEnforce;
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

/// <devdoc>
/// </devdoc>
public XmlReadMode ReadXml(XmlReader reader, XmlReadMode mode)
{
    return ReadXml(reader, mode, false);
}

internal XmlReadMode ReadXml(XmlReader reader, XmlReadMode mode, bool
denyResolving)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.ReadXml|INFO> %d#,
mode=%d{ds.XmlReadMode}, denyResolving=%d{bool}\n", ObjectID, (int)mode, denyResolving);
    try {

        XmlReadMode ret = mode;

        if (reader == null)
            return ret;

        if (mode == XmlReadMode.Auto) {
            // Dev11 915079: nested ReadXml calls on the same DataSet must be
done outside of RowDiffIdUsage scope
            return ReadXml(reader);
        }

        DataTable.DSRowDiffIdUsageSection rowDiffIdUsage = new
DataTable.DSRowDiffIdUsageSection();
        try {

            bool fSchemaFound = false;

```

```

    bool fDataFound = false;
    bool fIsXdr = false;
    int iCurrentDepth = -1;

    // Dev11 904428: prepare and cleanup rowDiffId hashtable
    rowDiffIdUsage.Prepare(this);

    if (reader is XmlTextReader)
        ((XmlTextReader)reader).WhitespaceHandling =
WhitespaceHandling.Significant;

    XmlDocument xdoc = new XmlDocument(); // we may need this to infer
the schema

    if ((mode != XmlReadMode.Fragment) && (reader.NodeType ==
XmlNodeType.Element))
        iCurrentDepth = reader.Depth;

    reader.MoveToContent();
    XmlDataLoader xmlload = null;

    if (reader.NodeType == XmlNodeType.Element) {
        XmlElement topNode = null;
        if (mode == XmlReadMode.Fragment) {
            xdoc.AppendChild(xdoc.CreateElement("ds_sqlXmlWrappER"));
            topNode = xdoc.DocumentElement;
        }
        else { //handle the top node
            if ((reader.LocalName == Keywords.DIFFGRAM) &&
(reader.NamespaceURI == Keywords.DFFNS)) {
                if ((mode == XmlReadMode.DiffGram) || (mode ==
XmlReadMode.IgnoreSchema)) {
                    this.ReadXmlDiffgram(reader);
                    // read the closing tag of the current element
                    ReadEndElement(reader);
                }
                else {
                    reader.Skip();
                }
                return ret;
            }

            if (reader.LocalName == Keywords.XDR_SCHEMA &&
reader.NamespaceURI == Keywords.XDRNS) {

```

```

        // load XDR schema and exit
        if ((mode != XmlReadMode.IgnoreSchema) && (mode !=
XmlReadMode.InferSchema) &&
            (mode != XmlReadMode.InferTypedSchema))
        {
            ReadXDRSchema(reader);
        }
        else {
            reader.Skip();
        }
        return ret; //since the top level element is a schema
return
    }

    if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI == Keywords.XSDNS) {
        // load XSD schema and exit
        if ((mode != XmlReadMode.IgnoreSchema) && (mode !=
XmlReadMode.InferSchema) &&
            (mode != XmlReadMode.InferTypedSchema))
        {
            ReadXSDSchema(reader, denyResolving);
        }
        else
            reader.Skip();
        return ret; //since the top level element is a schema
return
    }

    if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI.StartsWith(Keywords.XSD_NS_START, StringComparison.Ordinal))
        throw
ExceptionBuilder.DataSetUnsupportedSchema(Keywords.XSDNS);

    // now either the top level node is a table and we load it
through dataReader...

    // ... or backup the top node and all its attributes
    topNode = xdoc.CreateElement(reader.Prefix, reader.LocalName,
reader.NamespaceURI);

    if (reader.HasAttributes) {
        int attrCount = reader.AttributeCount;
        for (int i = 0; i < attrCount; i++) {
            reader.MoveToAttribute(i);
            if

```

```

(reader.NamespaceURI.Equals(Keywords.XSD_XMLNS_NS))
    topNode.SetAttribute(reader.Name,
reader.GetAttribute(i));

    else {
        XmlAttribute attr =
topNode.SetAttributeNode(reader.LocalName, reader.NamespaceURI);
        attr.Prefix = reader.Prefix;
        attr.Value = reader.GetAttribute(i);
    }
}
}
reader.Read();
}

while (MoveToElement(reader, iCurrentDepth)) {

    if (reader.LocalName == Keywords.XDR_SCHEMA &&
reader.NamespaceURI == Keywords.XDRNS) {
        // load XDR schema
        if (!fSchemaFound && !fDataFound && (mode !=
XmlReadMode.IgnoreSchema) && (mode != XmlReadMode.InferSchema) &&
(mode != XmlReadMode.InferTypedSchema))
        {
            ReadXDRSchema(reader);
            fSchemaFound = true;
            fIsXdr = true;
        }
        else {
            reader.Skip();
        }
        continue;
    }

    if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI == Keywords.XSDNS) {
        // load XSD schema and exit
        if ((mode != XmlReadMode.IgnoreSchema) && (mode !=
XmlReadMode.InferSchema) &&
(mode != XmlReadMode.InferTypedSchema))
        {
            ReadXSDSchema(reader, denyResolving);
            fSchemaFound = true;
        }
        else {

```

```

        reader.Skip();
    }
    continue;
}

    if ((reader.LocalName == Keywords.DIFFGRAM) &&
(reader.NamespaceURI == Keywords.DFFNS)) {
        if ((mode == XmlReadMode.DiffGram) || (mode ==
XmlReadMode.IgnoreSchema)) {

            this.ReadXmlDiffgram(reader);
            ret = XmlReadMode.DiffGram;
        }
        else {
            reader.Skip();
        }
        continue;
    }

    if (reader.LocalName == Keywords.XSD_SCHEMA &&
reader.NamespaceURI.StartsWith(Keywords.XSD_NS_START, StringComparison.Ordinal))
        throw
ExceptionBuilder.DataSetUnsupportedSchema(Keywords.XSDNS);

    if (mode == XmlReadMode.DiffGram) {
        reader.Skip();
        continue; // we do not read data in diffgram mode
    }

    // if we are here we found some data
    fDataFound = true;

    if (mode == XmlReadMode.InferSchema || mode ==
XmlReadMode.InferTypedSchema) { //save the node in DOM until the end;
        XmlNode node = xdoc.ReadNode(reader);
        topNode.AppendChild(node);
    }
    else {
        if (xmlload == null)
            xmlload = new XmlDataLoader(this, fIsXdr, topNode,
mode == XmlReadMode.IgnoreSchema);
        xmlload.LoadData(reader);
    }
} //end of the while

```

```

        // read the closing tag of the current element
        ReadEndElement(reader);

        // now top node contains the data part
        xdoc.AppendChild(topNode);
        if (xmlload == null)
            xmlload = new XmlDataLoader(this, fIsXdr, mode ==
XmlReadMode.IgnoreSchema);

        if (mode == XmlReadMode.DiffGram) {
            // we already got the diffs through XmlReader interface
            return ret;
        }

        // Load Data
        if (mode == XmlReadMode.InferSchema || mode ==
XmlReadMode.InferTypedSchema) {
            InferSchema(xdoc, null, mode);
            ret = XmlReadMode.InferSchema;
            xmlload.FromInference = true;
            //
        }
        try {
            xmlload.LoadData(xdoc);
        }
        finally {
            xmlload.FromInference = false;
        }
    }

    return ret;
}
finally {
    // Dev11 904428: prepare and cleanup rowDiffId hashtable
    rowDiffIdUsage.Cleanup();
}
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

```

```

/// <devdoc>

```

```

/// </devdoc>
public XmlReadMode ReadXml(Stream stream, XmlReadMode mode)
{
    if (stream == null)
        return XmlReadMode.Auto;

    XmlTextReader reader = (mode == XmlReadMode.Fragment) ? new
XmlTextReader(stream, XmlNodeType.Element, null) : new XmlTextReader(stream);
    // Prevent Dtd entity in dataset
    reader.XmlResolver = null;
    return ReadXml(reader, mode, false);
}

/// <devdoc>
/// </devdoc>
public XmlReadMode ReadXml(TextReader reader, XmlReadMode mode)
{
    if (reader == null)
        return XmlReadMode.Auto;

    XmlTextReader xmlreader = (mode == XmlReadMode.Fragment) ? new
XmlTextReader(reader.ReadToEnd(), XmlNodeType.Element, null) : new XmlTextReader(reader);
    // Prevent Dtd entity in dataset
    xmlreader.XmlResolver = null;
    return ReadXml(xmlreader, mode, false);
}

/// <devdoc>
/// </devdoc>
[ResourceExposure(ResourceScope.Machine)]
[ResourceConsumption(ResourceScope.Machine)]
public XmlReadMode ReadXml(string fileName, XmlReadMode mode)
{
    XmlTextReader xr = null;
    if (mode == XmlReadMode.Fragment) {
        FileStream stream = new FileStream(fileName, FileMode.Open);
        xr = new XmlTextReader(stream, XmlNodeType.Element, null);
    }
    else
        xr = new XmlTextReader(fileName);

    // Prevent Dtd entity in dataset
    xr.XmlResolver = null;

```

```

        try
        {
            return ReadXml(xr, mode, false);
        }
        finally {
            xr.Close();
        }
    }

    /// <devdoc>
    ///     Writes schema and data for the DataSet.
    /// </devdoc>
    public void WriteXml(Stream stream)
    {
        WriteXml(stream, XmlWriteMode.IgnoreSchema);
    }

    /// <devdoc>
    /// </devdoc>
    public void WriteXml(TextWriter writer)
    {
        WriteXml(writer, XmlWriteMode.IgnoreSchema);
    }

    /// <devdoc>
    /// </devdoc>
    public void WriteXml(XmlWriter writer)
    {
        WriteXml(writer, XmlWriteMode.IgnoreSchema);
    }

    /// <devdoc>
    ///     <para>[To be supplied.]</para>
    /// </devdoc>
    [ResourceExposure(ResourceScope.Machine)]
    [ResourceConsumption(ResourceScope.Machine)]
    public void WriteXml(String fileName)
    {
        WriteXml(fileName, XmlWriteMode.IgnoreSchema);
    }

    /// <devdoc>
    ///     Writes schema and data for the DataSet.

```

```

/// </devdoc>
public void WriteXml(Stream stream, XmlWriteMode mode)
{
    if (stream != null) {
        XmlTextWriter w = new XmlTextWriter(stream, null);
        w.Formatting = Formatting.Indented;

        WriteXml(w, mode);
    }
}

/// <devdoc>
/// </devdoc>
public void WriteXml(TextWriter writer, XmlWriteMode mode)
{
    if (writer != null) {
        XmlTextWriter w = new XmlTextWriter(writer);
        w.Formatting = Formatting.Indented;

        WriteXml(w, mode);
    }
}

/// <devdoc>
/// </devdoc>
public void WriteXml(XmlWriter writer, XmlWriteMode mode)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.WriteXml|API> %d#",
mode=%d{ds.XmlWriteMode}\n", ObjectID, (int)mode);
    try {
        // Generate SchemaTree and write it out
        if (writer != null) {

            if (mode == XmlWriteMode.DiffGram) {
                // Create and save the updates
                // new UpdateTreeGen(UpdateTreeGen.UPDATE,
(DataRowState)(-1), this).Save(writer, null);
                new NewDiffgramGen(this).Save(writer);
            }
            else {
                // Create and save xml data
                new XmlDataTreeWriter(this).Save(writer, mode ==
XmlWriteMode.WriteSchema);

```

```

        }
    }
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

/// <devdoc>
///     <para>[To be supplied.]</para>
/// </devdoc>
[ResourceExposure(ResourceScope.Machine)]
[ResourceConsumption(ResourceScope.Machine)]
public void WriteXml(String fileName, XmlWriteMode mode)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.WriteXml|API> %d#, fileName='%ls',
mode=%d{ds.XmlWriteMode}\n", ObjectID, fileName, (int)mode);
    XmlTextWriter xw = new XmlTextWriter(fileName, null);
    try {
        xw.Formatting = Formatting.Indented;
        xw.WriteStartDocument(true);
        if (xw != null) {
            // Create and save the updates
            if (mode == XmlWriteMode.DiffGram) {
                new NewDiffgramGen(this).Save(xw);
            }
            else {
                // Create and save xml data
                new XmlDataTreeWriter(this).Save(xw, mode ==
XmlWriteMode.WriteSchema);
            }
        }
        xw.WriteEndDocument();
    }
    finally {
        xw.Close();
        Bid.ScopeLeave(ref hscp);
    }
}

/// <devdoc>
///     <para>
///         Gets the collection of parent relations which belong to a

```

```

    ///      specified table.
    ///      </para>
    /// </devdoc>
    internal DataRelationCollection GetParentRelations(DataTable table)
    {
        return table.ParentRelations;
    }

    /// <devdoc>
    ///      <para>
    ///          Merges this <see cref='System.Data.DataSet'/> into a specified <see
    cref='System.Data.DataSet'/>.
    ///      </para>
    /// </devdoc>
    public void Merge(DataSet dataSet)
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.Merge|API> %d#, dataSet=%d\n",
ObjectID, (dataSet != null) ? dataSet.ObjectID : 0);
        try {
            Merge(dataSet, false, MissingSchemaAction.Add);
        }
        finally {
            Bid.ScopeLeave(ref hscp);
        }
    }

    /// <devdoc>
    ///      <para>
    ///          Merges this <see cref='System.Data.DataSet'/> into a specified <see
    cref='System.Data.DataSet'/> preserving changes according to
    ///          the specified argument.
    ///      </para>
    /// </devdoc>
    public void Merge(DataSet dataSet, bool preserveChanges)
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.Merge|API> %d#, dataSet=%d,
preserveChanges=%d{bool}\n", ObjectID, (dataSet != null) ? dataSet.ObjectID : 0,
preserveChanges);
        try {
            Merge(dataSet, preserveChanges, MissingSchemaAction.Add);
        }
        finally {

```

```

        Bid.ScopeLeave(ref hscp);
    }
}

/// <devdoc>
///     <para>
///         Merges this <see cref='System.Data.DataSet' /> into a specified <see
cref='System.Data.DataSet' /> preserving changes according to
///         the specified argument, and handling an incompatible schema according
to the
///         specified argument.
///     </para>
/// </devdoc>
public void Merge(DataSet dataSet, bool preserveChanges, MissingSchemaAction
missingSchemaAction) {
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.Merge|API> %d#, dataSet=%d,
preserveChanges=%d{bool}, missingSchemaAction=%d{ds.MissingSchemaAction}\n", ObjectID,
(dataSet != null) ? dataSet.ObjectID : 0, preserveChanges, (int)missingSchemaAction);
    try {

        // Argument checks
        if (dataSet == null)
            throw ExceptionBuilder.ArgumentNull("dataSet");

        switch (missingSchemaAction) { // @perfnote: Enum.IsDefined
            case MissingSchemaAction.Add:
            case MissingSchemaAction.Ignore:
            case MissingSchemaAction.Error:
            case MissingSchemaAction.AddWithKey:
                Merger merger = new Merger(this, preserveChanges,
missingSchemaAction);
                merger.MergeDataSet(dataSet);
                break;
            default:
                throw Common.ADP.InvalidMissingSchemaAction(missingSchemaAction);
        }
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

/// <devdoc>

```

```

    /// <para>
    ///     Merges this <see cref='System.Data.DataTable'/> into a specified <see
    cref='System.Data.DataTable'/>.
    /// </para>
    /// </devdoc>
    public void Merge(DataTable table)
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.Merge|API> %d#, table=%d\n", ObjectID,
(table != null) ? table.ObjectID : 0);
        try {
            Merge(table, false, MissingSchemaAction.Add);
        }
        finally {
            Bid.ScopeLeave(ref hscp);
        }
    }

    /// <devdoc>
    /// <para>
    ///     Merges this <see cref='System.Data.DataTable'/> into a specified <see
    cref='System.Data.DataTable'/>. with a value to preserve changes
    ///     made to the target, and a value to deal with missing schemas.
    /// </para>
    /// </devdoc>
    public void Merge(DataTable table, bool preserveChanges, MissingSchemaAction
missingSchemaAction)
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.Merge|API> %d#, table=%d,
preserveChanges=%d{bool}, missingSchemaAction=%d{ds.MissingSchemaAction}\n", ObjectID,
(table != null) ? table.ObjectID : 0, preserveChanges, (int)missingSchemaAction);
        try {
            // Argument checks
            if (table == null)
                throw ExceptionBuilder.ArgumentNull("table");

            switch (missingSchemaAction) { // @perfnote: Enum.IsDefined
                case MissingSchemaAction.Add:
                case MissingSchemaAction.Ignore:
                case MissingSchemaAction.Error:
                case MissingSchemaAction.AddWithKey:
                    Merger merger = new Merger(this, preserveChanges,
missingSchemaAction);

```

```

        merger.MergeTable(table);
        break;
    default:
        throw Common.ADP.InvalidMissingSchemaAction(missingSchemaAction);
    }
}

finally {
    Bid.ScopeLeave(ref hscp);
}
}

/// <devdoc>
///     <para>[To be supplied.]</para>
/// </devdoc>
public void Merge(DataRow[] rows)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.Merge|API> %d#, rows\n", ObjectID);
    try {
        Merge(rows, false, MissingSchemaAction.Add);
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

/// <devdoc>
///     <para>[To be supplied.]</para>
/// </devdoc>
public void Merge(DataRow[] rows, bool preserveChanges, MissingSchemaAction
missingSchemaAction)
{
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.Merge|API> %d#,
preserveChanges=%d{bool}, missingSchemaAction=%d{ds.MissingSchemaAction}\n", ObjectID,
preserveChanges, (int)missingSchemaAction);
    try {
        // Argument checks
        if (rows == null)
            throw ExceptionBuilder.ArgumentNull("rows");

        switch (missingSchemaAction) { // @perfnote: Enum.IsDefined
            case MissingSchemaAction.Add:

```

```

        case MissingSchemaAction.Ignore:
        case MissingSchemaAction.Error:
        case MissingSchemaAction.AddWithKey:
            Merger merger = new Merger(this, preserveChanges,
missingSchemaAction);
            merger.MergeRows(rows);
            break;
        default:
            throw Common.ADP.InvalidMissingSchemaAction(missingSchemaAction);
    }
}
finally {
    Bid.ScopeLeave(ref hscp);
}
}

protected virtual void OnPropertyChanging(PropertyChangedEventArgs pcevent)
{
    if (onPropertyChangingDelegate != null)
        onPropertyChangingDelegate(this, pcevent);
}

/// <devdoc>
///     Inheriting classes should override this method to handle this event.
///     Call base.OnMergeFailed to send this event to any registered event
///     listeners.
/// </devdoc>
internal void OnMergeFailed(MergeFailedEventArgs mfevent)
{
    if (onMergeFailed != null)
        onMergeFailed(this, mfevent);
    else
        throw ExceptionBuilder.MergeFailed(mfevent.Conflict);
}

internal void RaiseMergeFailed(DataTable table, string conflict,
MissingSchemaAction missingSchemaAction)
{
    if (MissingSchemaAction.Error == missingSchemaAction)
        throw ExceptionBuilder.MergeFailed(conflict);

    MergeFailedEventArgs mfevent = new MergeFailedEventArgs(table, conflict);
    OnMergeFailed(mfevent);
    return;
}

```

```

    }

    internal void OnDataRowCreated(DataRow row) {
        if (onDataRowCreated != null)
            onDataRowCreated(this, row);
    }

    internal void OnClearFunctionCalled(DataTable table) {
        if (onClearFunctionCalled != null)
            onClearFunctionCalled(this, table);
    }

    private void OnInitialized() {
        if (onInitialized != null) {
            onInitialized(this, EventArgs.Empty);
        }
    }

    /// <devdoc>
    /// This method should be overridden by subclasses to restrict tables being
removed.
    /// </devdoc>
    protected internal virtual void OnRemoveTable(DataTable table) {
    }

    internal void OnRemovedTable(DataTable table) {
        DataViewManager viewManager = defaultManager;
        if (null != viewManager) {
            viewManager.DataViewSettings.Remove(table);
        }
    }

    /// <devdoc>
    /// This method should be overridden by subclasses to restrict tables being
removed.
    /// </devdoc>
    protected virtual void OnRemoveRelation(DataRelation relation)
    {
    }

    //
    internal void OnRemoveRelationHack(DataRelation relation)
    {
        OnRemoveRelation(relation);
    }

```

```

}

protected internal void RaisePropertyChanging(string name)
{
    OnPropertyChanging(new PropertyChangedEventArgs(name));
}

internal DataTable[] TopLevelTables()
{
    return TopLevelTables(false);
}

internal DataTable[] TopLevelTables(bool forSchema)
{
    // first let's figure out if we can represent the given dataSet as a tree
using
    // the fact that all connected undirected graphs with n-1 edges are trees.
    List<DataTable> topTables = new List<DataTable>();

    if (forSchema) {
        // prepend the tables that are nested more than once
        for (int i = 0; i < Tables.Count; i++)
        {
            DataTable table = Tables[i];
            if (table.NestedParentsCount > 1 || table.SelfNested)
                topTables.Add(table);
        }
    }
    for (int i = 0; i < Tables.Count; i++)
    {
        DataTable table = Tables[i];
        if (table.NestedParentsCount == 0 && !topTables.Contains(table))
            topTables.Add(table);
    }
    if (topTables.Count == 0)
        return zeroTables;
    return topTables.ToArray();
}

/// <devdoc>
/// This method rolls back all the changes to have been made to this DataSet
since
    /// it was loaded or the last time AcceptChanges was called.

```

```

    /// Any rows still in edit-mode cancel their edits. New rows get removed.
    Modified and
    /// Deleted rows return back to their original state.
    /// </devdoc>
    public virtual void RejectChanges()
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.RejectChanges|API> %d#\n", ObjectID);
        try {
            bool fEnforce = EnforceConstraints;
            EnforceConstraints = false;
            for (int i = 0; i < Tables.Count; i++)
                Tables[i].RejectChanges();
            EnforceConstraints = fEnforce;
        }
        finally {
            Bid.ScopeLeave(ref hscp);
        }
    }

    /// <devdoc>
    ///     Resets the dataSet back to it's original state. Subclasses should
    override
    ///     to restore back to it's original state.
    ///
    public virtual void Reset()
    {
        IntPtr hscp;
        Bid.ScopeEnter(out hscp, "<ds.DataSet.Reset|API> %d#\n", ObjectID);
        try {
            for (int i = 0; i < Tables.Count; i++) {
                ConstraintCollection cons = Tables[i].Constraints;
                for (int j = 0; j < cons.Count; ) {
                    if (cons[j] is ForeignKeyConstraint) {
                        cons.Remove(cons[j]);
                    }
                    else
                        j++;
                }
            }
            // SQLBU 502734: because of SQLBU 501916, dependent tables need to be
            notified when a table is cleared
            // if relations are removed first, then exceptions happen

```

```

        Clear();
        Relations.Clear();
        Tables.Clear();
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

internal bool ValidateCaseConstraint() {
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.ValidateCaseConstraint|INFO> %d#\n",
ObjectID);
    try {
        DataRelation relation = null;
        for (int i = 0; i < Relations.Count; i++) {
            relation = Relations[i];
            if (relation.ChildTable.CaseSensitive !=
relation.ParentTable.CaseSensitive)
                return false;
        }

        ForeignKeyConstraint constraint = null;
        ConstraintCollection constraints = null;
        for (int i = 0; i < Tables.Count; i++) {
            constraints = Tables[i].Constraints;
            for (int j = 0; j < constraints.Count; j++) {
                if (constraints[j] is ForeignKeyConstraint) {
                    constraint = (ForeignKeyConstraint)constraints[j];
                    if (constraint.Table.CaseSensitive !=
constraint.RelatedTable.CaseSensitive)
                        return false;
                }
            }
        }
        return true;
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

internal bool ValidateLocaleConstraint() {
    IntPtr hscp;

```

```

        Bid.ScopeEnter(out hscp, "<ds.DataSet.ValidateLocaleConstraint|INFO> %d#\n",
ObjectID);
        try {
            DataRelation relation = null;
            for (int i = 0; i < Relations.Count; i++) {
                relation = Relations[i];
                if (relation.ChildTable.Locale.LCID !=
relation.ParentTable.Locale.LCID)
                    return false;
            }

            ForeignKeyConstraint constraint = null;
            ConstraintCollection constraints = null;
            for (int i = 0; i < Tables.Count; i++) {
                constraints = Tables[i].Constraints;
                for (int j = 0; j < constraints.Count; j++) {
                    if (constraints[j] is ForeignKeyConstraint) {
                        constraint = (ForeignKeyConstraint)constraints[j];
                        if (constraint.Table.Locale.LCID !=
constraint.RelatedTable.Locale.LCID)
                            return false;
                    }
                }
            }
            return true;
        }
        finally {
            Bid.ScopeLeave(ref hscp);
        }
    }

    // Microsoft: may be better to rewrite this as nonrecursive?
    internal DataTable FindTable(DataTable baseTable, PropertyDescriptor[] props, int
propStart) {
        if (props.Length < propStart + 1)
            return baseTable;

        PropertyDescriptor currentProp = props[propStart];

        if (baseTable == null) {
            // the accessor is the table name. if we don't find it, return null.
            if (currentProp is DataTablePropertyDescriptor) {
                return FindTable(((DataTablePropertyDescriptor)currentProp).Table,
props, propStart + 1);
            }
        }
    }

```

```

    }
    return null;
}

    if (currentProp is DataRelationPropertyDescriptor) {
        return
FindTable(((DataRelationPropertyDescriptor)currentProp).Relation.ChildTable, props,
propStart + 1);
    }

    return null;
}

    protected virtual void ReadXmlSerializable(XmlReader reader) {
        //WebData 96421 and 104709, this is DataSet side fix for the thrown exception
        // <DataSet xsi:nil="true"> does not mean DataSet is null, but it does not
have any child
        // so dont do anything, ignore the attributes and just return empty DataSet;
        this.UseDataSetSchemaOnly = false;
        this.UdtIsWrapped = false;

        if (reader.HasAttributes) {
            const string xsinill = Keywords.XSI + ":" + Keywords.XSI_NIL;
            if (reader.MoveToAttribute(xsinill)) {
                string nilAttrib = reader.GetAttribute(xsinill);
                if (string.Compare(nilAttrib, "true", StringComparison.Ordinal) == 0)
// case sensitive true comparison
                    MoveToElement(reader, 1);
                    return;
            }
        }

        const string useDataSetSchemaOnlyString = Keywords.MSD + ":" +
Keywords.USEDATASETSHEMAONLY;
        if (reader.MoveToAttribute(useDataSetSchemaOnlyString)) {
            string _useDataSetSchemaOnly =
reader.GetAttribute(useDataSetSchemaOnlyString);
            if (string.Equals(_useDataSetSchemaOnly, "true",
StringComparison.Ordinal) ||
                string.Equals(_useDataSetSchemaOnly, "1",
StringComparison.Ordinal))
            {
                this.UseDataSetSchemaOnly = true;
            }
        }
    }
}

```

```

        else if (!string.Equals(_useDataSetSchemaOnly, "false",
StringComparison.Ordinal) &&
                !string.Equals(_useDataSetSchemaOnly, "0",
StringComparison.Ordinal))
        {
            throw
ExceptionBuilder.InvalidAttributeValue(Keywords.USEDATASETSHEMAONLY,
_useDataSetSchemaOnly);
        }
    }

    const string udtIsWrappedString = Keywords.MSD + ":" +
Keywords.UDTCOLUMNVALUEWRAPPED;
    if (reader.MoveToAttribute(udtIsWrappedString)) {
        string _udtIsWrappedString = reader.GetAttribute(udtIsWrappedString);
        if (string.Equals(_udtIsWrappedString, "true",
StringComparison.Ordinal) ||
            string.Equals(_udtIsWrappedString, "1",
StringComparison.Ordinal))
        {
            this.UdtIsWrapped = true;
        }
        else if (!string.Equals(_udtIsWrappedString, "false",
StringComparison.Ordinal) &&
                !string.Equals(_udtIsWrappedString, "0",
StringComparison.Ordinal))
        {
            throw
ExceptionBuilder.InvalidAttributeValue(Keywords.UDTCOLUMNVALUEWRAPPED,
_udtIsWrappedString);
        }
    }

    ReadXml(reader, XmlReadMode.DiffGram, true);
}

protected virtual System.Xml.Schema.XmlSchema GetSchemaSerializable() {
    return null;
}

/*****
To publish
int_tools\webserviceadmin.exe -install Application Application.dll
*****/

```

To test

`http://localhost/application/service.asmx?wsdl`

or

`wsdl.exe /namespace:UserNamespace http://localhost/Application/Service.asmx?wsdl /out:Service.cs`

The V1.0 & V1.1 WSDL for Untyped DataSet being returned as a result (no parameters)

```
<s:element name="anyUserSpecifiedMethodName">
  <!-- This is where parameters go -->
  <s:complexType />
</s:element>
<s:element name="anyUserSpecifiedMethodName"+"Response">
  <s:complexType>
    <s:sequence>
      <s:element minOccurs="0" maxOccurs="1"
name="anyUserSpecifiedMethodName"+"Result">
        <s:complexType>
          <s:sequence>
            <s:element ref="s:schema" />
            <s:any />
          </s:sequence>
        </s:complexType>
      </s:element>
    </s:sequence>
  </s:complexType>
</s:element>
```

The V1.0 & V1.1 WSDL for Untyped DataSet as parameter, string for the result

```
<s:element name="anyUserSpecifiedMethodName">
  <!-- This is where parameters go -->
  <s:complexType>
    <s:sequence>
      <s:element minOccurs="0" maxOccurs="1" name="set">
        <s:complexType>
          <s:sequence>
            <s:element ref="s:schema" />
            <s:any />
          </s:sequence>
        </s:complexType>
      </s:element>
    </s:sequence>
  </s:complexType>
```

```

</s:element>
<s:element name="anyUserSpecifiedMethodName"+"Response">
  <s:complexType>
    <s:sequence>
      <s:element minOccurs="0" maxOccurs="1"
name="anyUserSpecifiedMethodName"+"Result" type="s:string" />
    </s:sequence>
  </s:complexType>
</s:element>

```

The V2.0 WSDL for Untyped DataSet being returned as a result (no parameters)

```

<s:element name="anyUserSpecifiedMethodName">
  <!-- This is where parameters go -->
  <s:complexType />
</s:element>
<s:element name="anyUserSpecifiedMethodName"+"Response">
  <s:complexType>
    <s:sequence>
      <s:element minOccurs="0" maxOccurs="1"
name="anyUserSpecifiedMethodName"+"Result">
        <s:complexType>
          <s:sequence maxOccurs="unbounded">
            <s:any minOccurs="0"
namespace="http://www.w3.org/2001/XMLSchema" processContents="lax" />
            <s:any minOccurs="0" namespace="urn:schemas-microsoft-
com:xml-diffgram-v1" processContents="lax" />
          </s:sequence>
        </s:complexType>
      </s:element>
    </s:sequence>
  </s:complexType>
</s:element>

```

The V2.0 WSDL for Untyped DataSet as a parameter, string for the result

```

<s:element name="anyUserSpecifiedMethodName">
  <!-- This is where parameters go -->
  <s:complexType>
    <s:sequence>
      <s:element minOccurs="0" maxOccurs="1"
name="anyUserSpecifiedParameterName">
        <s:complexType>
          <s:sequence maxOccurs="unbounded">
            <s:any minOccurs="0"
namespace="http://www.w3.org/2001/XMLSchema" processContents="lax" />

```

```

        <s:any minOccurs="0" namespace="urn:schemas-microsoft-
com:xml-diffgram-v1" processContents="lax" />
    </s:sequence>
</s:complexType>
</s:element>
</s:sequence>
</s:complexType>
</s:element>
<s:element name="anyUserSpecifiedMethodName"+"Response">
    <s:complexType>
        <s:sequence>
            <s:element minOccurs="0" maxOccurs="1"
name="anyUserSpecifiedMethodName"+"Result" type="s:string" />
        </s:sequence>
    </s:complexType>
</s:element>

```

The V1.0, V1.1 & V2.0 WSDL for Typed DataSet

```

<s:import schemaLocation="http://localhost/application/service.asmx?
schema=typedDataSetName" namespace="typedDataSetName" />
<s:element name="anyUserSpecifiedMethodName">
    <!-- This is where parameters go -->
    <s:complexType />
</s:element>
<s:element name="anyUserSpecifiedMethodName"+"Response">
    <s:complexType>
        <s:sequence>
            <s:element minOccurs="0" maxOccurs="1"
name="anyUserSpecifiedMethodName"+"Result">
                <s:complexType>
                    <s:sequence>
                        <s:any namespace="typedDataSetName" />
                    </s:sequence>
                </s:complexType>
            </s:element>
        </s:sequence>
    </s:complexType>
</s:element>

```

*****/

```

public static XmlSchemaComplexType GetDataSetSchema(XmlSchemaSet schemaSet) {
    // For performance reasons we are exploiting the fact that config files
content is constant
    // for a given appdomain so we can safely cache the prepared schema complex
type and reuse it

```

```

        if (schemaTypeForWSDL == null) { // to change the config file, appdomain
needs to restart; so it seems safe to cache the schema
            XmlSchemaComplexType tempWSDL = new XmlSchemaComplexType();
            XmlSchemaSequence sequence = new XmlSchemaSequence();

            if (PublishLegacyWSDL())
            { // Default is Version 1.0
                XmlSchemaElement elem = new XmlSchemaElement();
                elem.RefName = new XmlQualifiedName(Keywords.XSD_SCHEMA,
Keywords.XSDNS);

                sequence.Items.Add(elem);
                XmlSchemaAny any = new XmlSchemaAny();
                sequence.Items.Add(any);
            }
            else
            { // this means Version == V2.0 or newer
                XmlSchemaAny any = new XmlSchemaAny();
                any.Namespace = XmlSchema.Namespace;
                any.MinOccurs = 0;
                any.ProcessContents = XmlSchemaContentProcessing.Lax;
                sequence.Items.Add(any);

                any = new XmlSchemaAny();
                any.Namespace = Keywords.DFFNS;
                any.MinOccurs = 0; // when recognizing WSDL - MinOccurs="0" denotes
DataSet, a MinOccurs="1" for DataTable
                any.ProcessContents = XmlSchemaContentProcessing.Lax;
                sequence.Items.Add(any);
                sequence.MaxOccurs = Decimal.MaxValue;
            }
            tempWSDL.Particle = sequence;

            schemaTypeForWSDL = tempWSDL;
        }
        return schemaTypeForWSDL;
    }

    /*****
<configuration>
  <configSections>
    <section name="system.data.dataset"
type="System.Configuration.NameValueFileSectionHandler, System,
Version=%ASSEMBLY_VERSION%, Culture=neutral, PublicKeyToken=%ECMA_PUBLICKEY%"
restartOnExternalChanges="false" />

```

```

</configSections>
<system.data.dataset>
    <!-- if WSDL_VERSION is missing it will default to 1.0
        <add key="WSDL_VERSION" value="1.0"/>
    -->
    <add key="WSDL_VERSION" value="2.0"/>
</system.data.dataset>
</configuration>
*****/

private static bool PublishLegacyWSDL()
{
    Single version = 1.0f; // Default is Version 1.0
    NameValueCollection settings =
(NameValueCollection)PrivilegedConfigurationManager.GetSection(Keywords.WS_DATASETFULLQNA
ME);

    if (settings != null)
    {
        string[] values = settings.GetValues(Keywords.WS_VERSION);
        if ((null != values) && (0 < values.Length) && (null != values[0]))
        { // will throw FormatException if not a valid number
            version = Single.Parse(values[0], CultureInfo.InvariantCulture);
        }
    }

    return (version < 2.0f); // if config does not exist, Default is Version 1.0
}

XmlSchema IXmlSerializable.GetSchema() {
    if (GetType() == typeof(DataSet)) {
        return null;
    }
    MemoryStream stream = new MemoryStream();
    // WriteXmlSchema(new XmlTextWriter(stream, null));
    XmlWriter writer = new XmlTextWriter(stream, null);
    if (writer != null) {
        (new XmlTreeGen(SchemaFormat.WebService)).Save(this, writer);
    }
    stream.Position = 0;
    return XmlSchema.Read(new XmlTextReader(stream), null);
    // return GetSchemaSerializable();
}

void IXmlSerializable.ReadXml(XmlReader reader) {
    bool fNormalization = true;
    XmlTextReader xmlTextReader = null;

```

```

IXmlTextParser xmlTextParser = reader as IXmlTextParser;
if (xmlTextParser != null) {
    fNormalization = xmlTextParser.Normalized;
    xmlTextParser.Normalized = false;
}
else {
    xmlTextReader = reader as XmlTextReader;
    if (xmlTextReader != null) {
        fNormalization = xmlTextReader.Normalization;
        xmlTextReader.Normalization = false;
    }
}

ReadXmlSerializable(reader);

if (xmlTextParser != null)
    xmlTextParser.Normalized = fNormalization;
else if (xmlTextReader != null)
    xmlTextReader.Normalization = fNormalization;
}

void IXmlSerializable.WriteXml(XmlWriter writer) {
    this.WriteXmlSchema(writer, SchemaFormat.WebService, (Converter<Type,
string>) null);
    this.WriteXml(writer, XmlWriteMode.DiffGram);
}

public virtual void Load(IDataReader reader, LoadOption loadOption,
FillErrorEventHandler errorHandler, params DataTable[] tables) {
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.Load|API> reader,
loadOption=%d{ds.LoadOption}", (int)loadOption);
    try {
        foreach (DataTable dt in tables) {
            Common.ADP.CheckArgumentNull(dt, "tables");
            if (dt.DataSet != this) {
                throw ExceptionBuilder.TableNotInTheDataSet(dt.TableName);
            }
        }
        Common.LoadAdapter adapter = new Common.LoadAdapter();
        adapter.FillLoadOption = loadOption;
        adapter.MissingSchemaAction = MissingSchemaAction.AddWithKey;
        if (null != errorHandler) {
            adapter.FillError += errorHandler;
        }
    }
}

```

```

        }
        adapter.FillFromReader(tables, reader, 0, 0);

        if (!reader.IsClosed && !reader.NextResult()) { //
            reader.Close();
        }
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

public void Load(IDataReader reader, LoadOption loadOption, params DataTable[]
tables) {
    Load(reader, loadOption, null, tables);
}

public void Load(IDataReader reader, LoadOption loadOption, params string[]
tables) {
    Common.ADP.CheckArgumentNull(tables, "tables");
    DataTable[] dataTables = new DataTable[tables.Length];
    for (int i = 0; i < tables.Length; i++) {
        DataTable tempDT = this.Tables[tables[i]];
        if (null == tempDT) {
            tempDT = new DataTable(tables[i]);
            // fxcop: new DataTable should inherit the CaseSensitive, Locale,
Namespace from DataSet
            Tables.Add(tempDT);
        }
        dataTables[i] = tempDT;
    }
    Load(reader, loadOption, null, dataTables);
}

public DataTableReader CreateDataReader() {
    if (Tables.Count == 0)
        throw ExceptionBuilder.CannotCreateDataReaderOnEmptyDataSet();

    DataTable[] dataTables = new DataTable[Tables.Count];
    for (int i = 0; i < Tables.Count; i++) {
        dataTables[i] = Tables[i];
    }
    return CreateDataReader(dataTables);
}

```

```

public DataTableReader CreateDataReader(params DataTable[] dataTables) {
    IntPtr hscp;
    Bid.ScopeEnter(out hscp, "<ds.DataSet.GetDataReader|API> %d#\n", ObjectID);
    try {
        if (dataTables.Length == 0)
            throw ExceptionBuilder.DataTableReaderArgumentIsEmpty();

        for (int i = 0; i < dataTables.Length; i++) {
            if (dataTables[i] == null)
                throw ExceptionBuilder.ArgumentContainsNullValue();
        }

        return new DataTableReader(dataTables);
    }
    finally {
        Bid.ScopeLeave(ref hscp);
    }
}

internal string MainTableName {
    get {
        return mainTableName;
    }
    set {
        this.mainTableName = value;
    }
}

internal int ObjectID {
    get {
        return _objectID;
    }
}
}

```

```

public class DataSetSchemaImporterExtension : SchemaImporterExtension {
    // DataSetSchemaImporterExtension is used for WebServices, it is used to
    recognize the schema of DataSet within wsdl
    // If a non 2.0 enabled DataSetSchemaImporterExtension, wsdl will generate a
    classes that you can't cast to dataset / datatable

    Hashtable importedTypes = new Hashtable();

```

```

    /// <devdoc>
    ///     <para>[To be supplied.]</para>
    /// </devdoc>
    public override string ImportSchemaType(string name, string schemaNamespace,
    XmlSchemaObject context, XmlSchemas schemas, XmlSchemaImporter importer, CodeCompileUnit
    compileUnit, CodeNamespace mainNamespace, CodeGenerationOptions options, CodeDomProvider
    codeProvider) {
        IList values = schemas.GetSchemas(schemaNamespace);
        if (values.Count != 1) {
            return null;
        }
        XmlSchema schema = values[0] as XmlSchema;
        if (schema == null)
            return null;
        XmlSchemaType type = (XmlSchemaType)schema.SchemaTypes[new
    XmlQualifiedName(name, schemaNamespace)];
        return ImportSchemaType(type, context, schemas, importer, compileUnit,
    mainNamespace, options, codeProvider);
    }

    /// <devdoc>
    ///     <para>[To be supplied.]</para>
    /// </devdoc>
    public override string ImportSchemaType(XmlSchemaType type, XmlSchemaObject
    context, XmlSchemas schemas, XmlSchemaImporter importer, CodeCompileUnit compileUnit,
    CodeNamespace mainNamespace, CodeGenerationOptions options, CodeDomProvider codeProvider)
    {
        if (type == null) {
            return null;
        }
        if (importedTypes[type] != null) {
            mainNamespace.Imports.Add(new
    CodeNamespaceImport(typeof(DataSet).Namespace));
            compileUnit.ReferencedAssemblies.Add("System.Data.dll");
            return (string)importedTypes[type];
        }
        if (!(context is XmlSchemaElement))
            return null;

        XmlSchemaElement e = (XmlSchemaElement)context;

        // recognizing the following is important, but not as part of SQLPT
    120015394: Support WSI Compliant WSDL for DataSet

```

```

        // <xs:element name="NewDataSet" msdata:IsDataSet="true">
        // see also SQLBU 338644, 410965, 423446
        //if (IsDataSet(e))
        //{
        //    return GenerateTypedDataSet(e, schemas, mainNamespace,
compileUnit.ReferencedAssemblies, codeProvider);
        //}

        if (type is XmlSchemaComplexType) {
            XmlSchemaComplexType ct = (XmlSchemaComplexType)type;

            if (ct.Particle is XmlSchemaSequence)
            {
                XmlSchemaObjectCollection items =
((XmlSchemaSequence)ct.Particle).Items;
                if ((2 == items.Count) && (items[0] is XmlSchemaAny) && (items[1] is
XmlSchemaAny))
                {
                    XmlSchemaAny any0 = (XmlSchemaAny)items[0];
                    XmlSchemaAny any1 = (XmlSchemaAny)items[1];
                    if ((any0.Namespace == XmlSchema.Namespace) && (any1.Namespace ==
"urn:schemas-microsoft-com:xml-diffgram-v1"))
                    {
                        // new diffgram format
                        string ns = null;
                        foreach (XmlSchemaAttribute a in ct.Attributes)
                        {
                            if (a.Name == "namespace")
                            {
                                ns = a.FixedValue.Trim();
                                break;
                            }
                        }
                        bool isDataSet = false;

                        // check for DataSet or DataTable
                        if (((XmlSchemaSequence)ct.Particle).MaxOccurs ==
Decimal.MaxValue)
                        {
                            isDataSet = true;
                        }
                        else if (any0.MaxOccurs == Decimal.MaxValue)
                        {
                            isDataSet = false;
                        }
                    }
                }
            }
        }
    }
}

```

```

        else
        {
            return null;
        }

        if (ns == null)
        {
            //Un-Typed DataSet / DataTable
            string typeName = isDataSet ? typeof(DataSet).FullName :
typeof(DataTable).FullName;

            importedTypes.Add(type, typeName);
            mainNamespace.Imports.Add(new
CodeNamespaceImport(typeof(DataSet).Namespace));
            compileUnit.ReferencedAssemblies.Add("System.Data.dll");
            return typeName;
        }
        else
        {
            // Typed DataSet / DataTable
            foreach (XmlSchema schema in schemas.GetSchemas(ns))
            {
                if ((schema != null) && (schema.Id != null))
                {
                    XmlSchemaElement ds = FindDataSetElement(schema);
// implement FindDataTableElement(schema)
                    if (ds != null)
                    {
                        return ImportSchemaType(ds.SchemaType, ds,
schemas, importer, compileUnit, mainNamespace, options, codeProvider);
                    }
                    // else return null
                }
            }
            return null;
        }
    }
}

if (ct.Particle is XmlSchemaSequence || ct.Particle is XmlSchemaAll) {
    XmlSchemaObjectCollection items =
((XmlSchemaGroupBase)ct.Particle).Items;
    if (items.Count == 2) {
        if (!(items[0] is XmlSchemaElement && items[1] is XmlSchemaAny))
return null;

        XmlSchemaElement schema = (XmlSchemaElement)items[0];
        if (!(schema.RefName.Name == "schema" && schema.RefName.Namespace

```

```

== XmlSchema.Namespace)) return null;
        string typeName = typeof(DataSet).FullName;
        importedTypes.Add(type, typeName);
        mainNamespace.Imports.Add(new
CodeNamespaceImport(typeof(DataSet).Namespace));
        compileUnit.ReferencedAssemblies.Add("System.Data.dll");
        return typeName;
    }
    else if (1 == items.Count)
    {
        XmlSchemaAny any = items[0] as XmlSchemaAny;
        if ((null != any) &&
            (null != any.Namespace) &&
            (any.Namespace.IndexOfAny(new char[] { '#', ' ' }) < 0)) //
special syntax (##any, ##other, ...) or more than one Uri present
        {
            foreach (XmlSchema schema in
schemas.GetSchemas(any.Namespace))
            {
                if ((null != schema) &&
                    (null != schema.Id))
                {
                    XmlSchemaElement ds = FindDataSetElement(schema);
                    if (ds != null)
                    {
                        return ImportSchemaType(ds.SchemaType, ds,
schemas, importer, compileUnit, mainNamespace, options, codeProvider);
                    }
                    // else return null
                }
                // else return null
            }
        }
        // else return null
    }
}
return null;
}

```

```

internal XmlSchemaElement FindDataSetElement(XmlSchema schema) {
    foreach (XmlSchemaObject item in schema.Items) {
        if (item is XmlSchemaElement && IsDataSet((XmlSchemaElement)item)) {

```

```

        return (XmlSchemaElement)item;
    }
}
return null;
}

```

```

internal string GenerateTypedDataSet(XmlSchemaElement element, XmlSchemas
schemas, CodeNamespace codeNamespace, StringCollection references, CodeDomProvider
codeProvider) {

```

```

    if (element == null)
        return null;

    if (importedTypes[element.SchemaType] != null)
        return (string)importedTypes[element.SchemaType];

```

```

    IList values = schemas.GetSchemas(element.QualifiedName.Namespace);
    if (values.Count != 1) {
        return null;
    }
    XmlSchema schema = values[0] as XmlSchema;
    if (schema == null)
        return null;

```

```

    DataSet ds = new DataSet();

```

```

    //
    using (MemoryStream stream = new MemoryStream()) {
        schema.Write(stream);
        stream.Position = 0;
        ds.ReadXmlSchema(stream);
    }

```

```

#pragma warning disable 618 // ignore obsolete warning about TypedDataSetGenerator

```

```

    CodeTypeDeclaration dsClass = new TypedDataSetGenerator().GenerateCode(ds,
codeNamespace, codeProvider.CreateGenerator());

```

```

#pragma warning restore 618

```

```

    string typeName = dsClass.Name;
    importedTypes.Add(element.SchemaType, typeName);
    references.Add("System.Data.dll");
    return typeName;
}

```

```

internal static bool IsDataSet(XmlSchemaElement e) {
    if (e.UnhandledAttributes != null) {

```

```
        foreach (XmlAttribute a in e.UnhandledAttributes) {
            if (a.LocalName == "IsDataSet" && a.NamespaceURI == Keywords.MSDNS) {
                // currently the msdata:IsDataSet uses its own format for the
boolean values
                if (a.Value == "True" || a.Value == "true" || a.Value == "1")
return true;
            }
        }
        return false;
    }
}
```

Archived from <https://github.com/microsoft/referencesource/blob/main/System.Data/System/Data/DataSet.cs>.
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